Readme for reproducibility submission of paper ID 152

1. Hardware and Software Requirements for *ORANClaw*

The artifact leverages Docker containers to minimize environmental dependencies and facilitate testing on standard x86_64 hardware. Unlike hardware-based radio setups, this artifact utilizes RF simulation to ensure accessibility without requiring Software Defined Radios (SDRs). To facilitate testing, the evaluation machine already has all the system libraries and python dependencies required to run *ORANClaw*. However, instructions to build from source are included in Section 1.5.

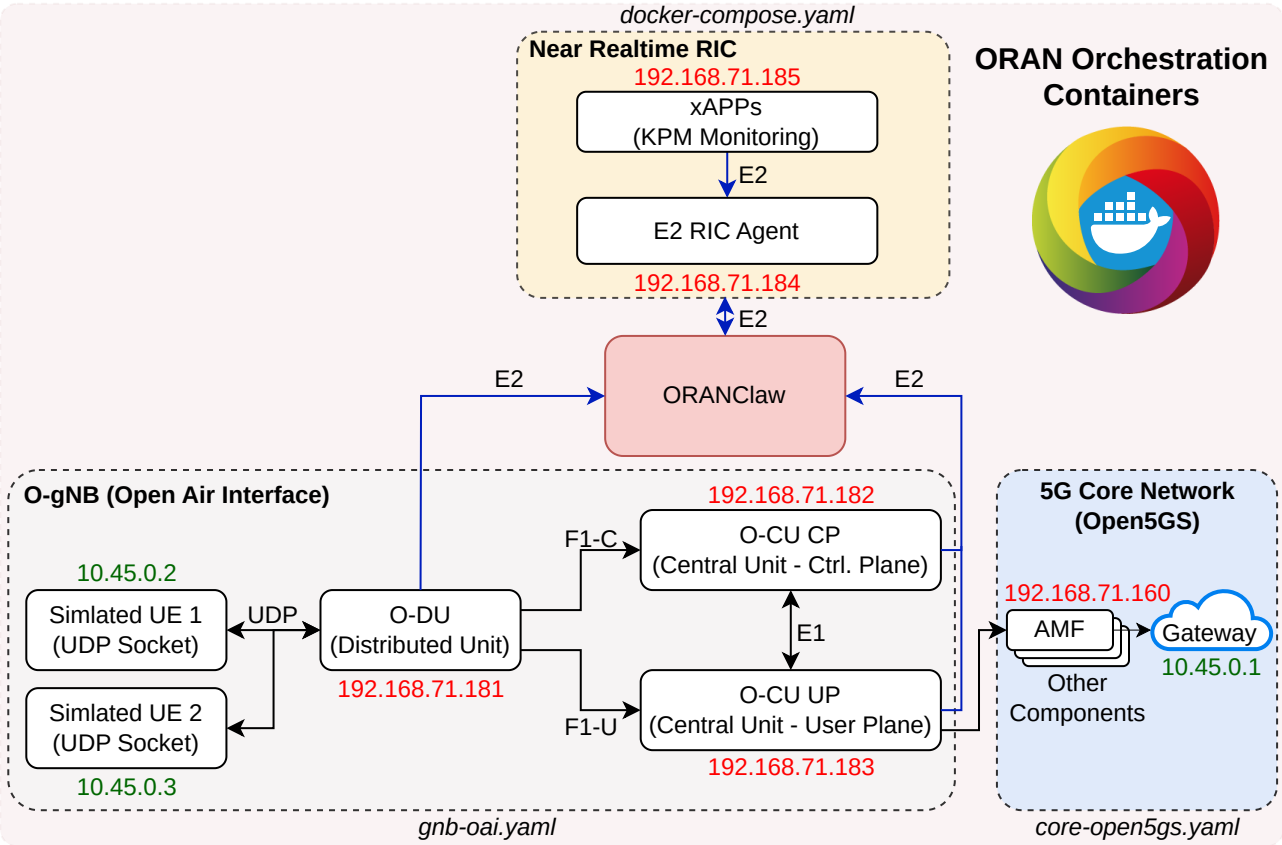

Figure 1: Overview of *ORANClaw* Evaluation Setup.

Figure 1 shows the dockerized environment used to deploy and test the system discussed in Section 4 of this paper. Specifically, it illustrates the OpenAirInterface implementation.

1.1. Hardware Dependencies

- **CPU:** Minimum 4 cores (8+ cores recommended for concurrent fuzzing).
- **RAM:** Minimum 16GB (32GB recommended for stable OAI gNB + UE simulation).
- **Storage:** At least 20GB of free space is required for Docker images, packet traces, and ns-3. If uncompressing the *ORANClaw* logs to generate plots for RQs (see 3.1), which require up to 500 GB, ensure that sufficient space is available.

1.2. Software Dependencies

The main runtime environment is based on Ubuntu 24.04. The following software must be installed on the host:

- **Docker:** Latest version for container orchestration.
- **Python $\geq$ 3.12:** Required for the MitM Fuzzing Engine (Included with artifact).
- **TShark $\geq$ 4.2.2:** For capturing and analyzing E2AP/SCTP traffic (Included with artifact).
- **asn1c compiler $\geq$ 7.8 license:** Required for encoding/decoding of ASN.1 messages (License included with artifact)

1.3. Project Structure

```
ORANClaw-E2-MitM-Fuzzing ..................Artifact repository for ORANClaw E2 MitM fuzzing framework
├── core-open5gs.yaml ......................................................Open5GS 5G core configuration
├── docker-compose.yaml ...............................Orchestration of O-RAN, RIC, and core components
├── Dockerfile ...............................................Container image for ORANClaw experiments
├── gnb-oai.yaml ..........................................................OAI gNB configuration (CU/DU)
├── requirements.sh ..............................................System-level dependency installation
├── configs ...................................................Configuration files for O-RAN components
│   ├── flexric.conf ..........................................................Near-RT RIC configuration
│   └── xapp.conf ................xApp configuration redirecting RIC IP (.129) to localhost to enable E2 MitM
├── asn1 ...........................................ASN.1 specifications, code generation, and fuzzing logic
│   ├── asn1files ...................................................E2AP and E2SM ASN.1 specifications
│   ├── asn1c ...........................................................ASN.1 compiler and runtime support
│   ├── src ...............................................Generated and E2AP/E2SM encoders and decoders
│   │   ├── reader.cpp ...................................................Customized encoder (PER-> JER)
│   │   └── reader_json.cpp ..............................................Customized decoder (JER-> PER)
│   ├── client_server.py ....................................ASN.1-aware fuzzing engine (OAI version)
│   ├── captures_bridge ....................................Capture files generated during fuzzing sessions
│   ├── container_logs ..............................OAI docker logs files generated during fuzzing sessions
│   ├── fuzzing_logs ......................................Fuzzer logs generated during fuzzingf sessions
│   ├── state_machines ...................................................Learned E2 protocol state machines
│   ├── state_machines_diff ........................Behavioral diffs between benign and fuzzed sessions
│   ├── fuzzing_logs ......................................Fuzzer logs generated during fuzzingf sessions
│   └── gen-code.sh ......................................Generate C/C++ code from ASN.1 files script
├── captures ...........................................PCAP traces from baseline and attack experiments
│   └── baseline.pcapng ...................................................Benign baseline capture file
├── libs ...................................................................Precompiled FlexRIC service models
│   └── flexric ....................................KPM, RC, MAC, RLC, PDCP, and TC service models
├── scripts ...................................................Experiment automation and UE management
│   ├── add_subscribers.sh ....................................Registers UEs in Open5GS (MongoDB)
│   └── run_ue_iperf.sh ...................................................Traffic generation for evaluation
├── docs ...................................................Design diagrams and system overview figures
├── vakt-ble-defender.zip ........... Pre-compiled binaries to create the state machines (wdissector libraries)
└── asn1c.zip ..............................................................Pre-compiled binaries of asn1c
```

1.4. How to Access

The remote evaluation machine is accessed exclusively via AnyDesk, which provides a graphical desktop interface for executing experiments and visualizing results in real-time.

Step 1: Install AnyDesk

```
# For Ubuntu/Debian
wget -qO - https://keys.anydesk.com/repos/DEB-GPG-KEY | sudo apt-key add -
echo "deb http://deb.anydesk.com/ all main" | sudo tee /etc/apt/sources.list.d/anydesk-stable.list
sudo apt update
sudo apt install anydesk
```

Alternatively, download the installer directly from https://anydesk.com/en/downloads/linux for your operating system (Windows, macOS, or Linux).

Step 2: Connect to Evaluation Machine
Launch AnyDesk and enter the connection details provided below (actual credentials will be shared with reviewers upon request):

```
AnyDesk Address: 123 456 789  # Example address
Password: artifact2025         # Example password
```

After successful authentication, you will have full desktop access to the evaluation machine with a pre-configured *ORANClaw* environment, including:

- All Docker containers ready to run
- Pre-installed fuzzing engine and dependencies
- Terminal access for running experiments
- GUI tools for monitoring and log visualization

NOTE: The actual AnyDesk address and password will be provided to artifact evaluation reviewers through a secure channel. The example credentials shown above (`123 456 789` and `artifact2025`) are for illustration purposes only.

1.5. Software Installation (build from source)

The installation of the *ORANClaw* testbed is facilitated through Docker. The following steps prepare the environment for fuzzing.
1 Clone **ORANClaw repository**.

```
cd $HOME
wget https://zenodo.org/records/18580510/files/ORANClaw-E2-MitM-Fuzzing.zip?download=1 -O ORANClaw-E2\
    -MitM-Fuzzing-AFFC.zip
mkdir ORANClaw-E2-MitM-Fuzzing-AFFC
unzip ORANClaw-E2-MitM-Fuzzing-AFFC.zip -d ORANClaw-E2-MitM-Fuzzing-AFFC
rm -rf ORANClaw-E2-MitM-Fuzzing-AFFC.zip
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC
```

2 **Install System Requirements**
This script will install all the necessary libraries, docker and docker composer.

```
chmod +x requirements.sh
./requirements.sh
source $HOME/.venvs/oran-env/bin/activate # activate virtual env
```

3 **Install *wdissector***
We leveraged the *wdissector* project to create the state machine components, `SM` and `SM DIFF`. However, this tool was originally compiled for Ubuntu 18.04.6 LTS (Bionic Beaver), making it difficult to compile on newer Ubuntu versions. For convenience, we provide a *.zip* file containing all the pre-compiled runtime libraries and binaries, which are invoked by Python from the fuzzing engine. To use it, simply uncompress the archive:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC
wget -O vakt-ble-defender.zip https://zenodo.org/records/18368683/files/vakt-ble-defender.zip?download=1
unzip vakt-ble-defender
rm -rf vakt-ble-defender.zip
```

4 Install **asn1c compiler license**.

For *ORANClaw* to encode/decode messages, is required to use asn1c compiler. For simplicity, we also provide a compiled version of E2AP V2.03 working with FlexRIC and OAI.

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/asn1
wget https://zenodo.org/records/18368683/files/asn1c.zip?download=1 -O asn1c.zip
unzip asn1c.zip
rm -rf asn1c.zip
chmod +x ./reader
chmod +x ./reader_json
```

Building from source, make sure to download asn1c compiler and after license purchasing type the lickey as folows:

```
# add license
./asn1/asn1c/bin/asn1c --lickey <key>
```

We do not include build-from-source instructions for these components, as they are maintained in external repositories and come with specific environment and licensing requirements. Instead, for ease of use, we provide pre-compiled binaries in compressed form that can be used in a plug-and-play manner. This avoids the need to build environments (e.g., Ubuntu 18.04 for *wdissector*) or manual dependency setup, while still supporting all the functionality required by *ORANClaw*.

5 Add custom ASN.1 versions (e.g., E2AP, E2SM-KPM).

Note that by default the project is compiled with *E2AP v2.03*, *E2SM-RC v1.03* and *E2SM-KPM v2.03*. However, *ORANClaw* is designed to be specification-agnostic. You can download different versions of E2AP (e.g., E2AP v3.0.0, E2AP-KPM v6.0.0) into the */asn1files* directory. To use a different version, simply modify the *gen-code.sh* script by updating the filename. An example of the scrip contents is shown in Listing 1.

Listing 1: *gen-code.sh* script

```
#gen-code.sh file:
./asn1c/bin/asn1c \
asn1files/E2SM-KPM-v05.00.asn \    ##Add your file name here
asn1files/asn_flexric/e2sm_rc_v1_03_modified.asn \  ##Add your file name here
asn1files/asn_flexric/e42ap_v2_03.asn -genTest \  ##Add your file name here
-per -json -w32 -server -depends -events -genPrtToStr \
-list -reader -trace -c++ -print -prtfmt details \
-srcdir src -make Makefile -w32 -table-unions -stream -pdu all
```

At this point, all components of *ORANClaw* are installed and ready for experimental evaluation. The testbed includes a fully configured Docker environment, the ASN.1 compiler with proper licensing, pre-compiled wdissector libraries for state machine analysis, and all necessary dependencies. You can now proceed to the experimental workflow described in the following sections to reproduce the results presented in the paper.

2. Experimental Evaluation

2.1. Evaluating *ORANClaw* with OpenAirInterface

1 Start Core Network

Run the command below in a separate terminal and wait until you see successful UDR related logs:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC
docker compose --profile core up # Terminal 1 - Core Network
```

Listing 2 shows successful UDR logs.

Listing 2: Core Network example output

```
udr-1    | 04/12 07:58:12.304: [sbi] INFO: [6250a1de-f8a2-41ee-bdd9-0f6b7e1be1b5] NF registered [\
    Heartbeat:10s] (../lib/sbi/nf-sm.c:221)
udm-1    | 04/12 07:58:12.304: [sbi] INFO: (NRF-notify) NF registered [6250a1de-f8a2-41ee-bdd9-0f6b7e1be\
    1b5:1] (../lib/sbi/nnrf-handler.c:924)
udm-1    | 04/12 07:58:12.304: [sbi] INFO: [UDR] (NRF-notify) NF Profile updated [6250a1de-f8a2-41ee-bdd\
    9-0f6b7e1be1b5:1] (../lib/sbi/nnrf-handler.c:938)
nrf-1    | 04/12 07:58:12.304: [nrf] INFO: [68e13194-f8a2-41ee-a707-db685e0fa5ce] Subscription created \
    until 2024-04-13T07:58:12.304328+00:00 [validity_duration:86400] (../src/nrf/nnrf-handler.c:445)
udr-1    | 04/12 07:58:12.304: [sbi] INFO: [68e13194-f8a2-41ee-a707-db685e0fa5ce] Subscription created \
    until 2024-04-13T07:58:12.304328+00:00 [duration:86400,validity:86399.999815,patch:43199.999907] \
    (../lib/sbi/nnrf-handler.c:708)
```

2 Add UE Subscribers

Add sample UE subscribers to the core network so tha the UE Simulator can register to the network:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/scripts/
./add_subcribers.sh
```

A successful output when subscribing the UEs is shown on Listing 3:

Listing 3: UE Registration output example

```
{
  acknowledged: true,
  insertedId: ObjectId('6618ec5e2d06d1bb877b2da9')
}
Done!
```

Note that you can modify the UE SIM card parameters (*IMSI, key, OPc, APN*) in the files *gnb-oai.yaml* and *scripts/add_subscribers.sh*.

③ Start the O-RAN gNB, UE simulation, and Near-Real-Time RIC
Run the command below in a separate terminal and wait until you observe successful UE-related logs:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
docker compose --profile gnb-rfsim up # Terminal 2 - gNB + UE Simulation + Near Realtime RIC
```

Successful UE logs:

Listing 4: UE Logs example

```
oai-ue-rfsimu-2-1  | [NR_PHY]   ============================================
oai-ue-rfsimu-2-1  | [NR_PHY]   Harq round stats for Downlink: 2735/1/0
oai-ue-rfsimu-2-1  | [NR_PHY]   ============================================
```

Note that two simulated UEs are started (containers `oai-ue-rfsimu-1-1` and `oai-ue-rfsimu-2-1`). You can add or remove UEs from the simulation by modifying file *gnb-oai.yaml*.

④ Run commands in the UE (iperf)
UE to Core Network
You can test UE Uplink/Downlink transfer speed by running iperf against the core network, which has default IP address of 10.45.0.1:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
docker compose exec -it oai-ue-rfsimu-1 iperf3 -c 10.45.0.1 -t0 # Terminal 3 - UE to Core Transfer
```

If the command is successful and the UE is registered to the core network, iperf should indicate a bitrate of about 100mbits/sec as shown below:

Listing 5: UE Uplink/Downlink output example

```
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-1.65   sec  22.2 MBytes   113 Mbits/sec    0             sender
```

Core Network to UE
Similarly, you can run iperf against UEs, which are usually registered to IP addresses 10.45.0.2 or 10.45.0.3:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
docker compose exec -it upf iperf3 -c 10.45.0.3 -t0 # Terminal 3 - Core to UE Transfer
```

2.2. Running *ORANClaw* (OAI)

① Start the Fuzzer
We start the *ORANClaw* fuzzer by running the following commands. The IP addresses of the Near-RT RIC, the xApp, and the MiTM proxy are defined in the script at lines *21–23*.

Listing 6: client_server.py script

```
IP_MITM = "192.168.71.129"
IP_XAPP = "192.168.71.185"
IP_RIC = "192.168.71.184"
```

To launch the fuzzer, run the following commands:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/asn1
python3 client_server.py
```

Once started, the fuzzer initializes the SCTP MiTM proxy and waits for an incoming xApp connection. An example of the expected output is shown in Listing 7.

Listing 7: *ORANClaw* SCTP MITM Proxy Output

```
==================================================
            SCTP MITM Proxy - Starting
==================================================
[+] Started tshark capture
```

```
[Genetic Algorithm] Enabled
[+] Connected to RIC at 192.168.71.184:36422
[*] Listening for xApp on 192.168.71.129:36422...
[*] Fuzzing enabled: True (Probability: 20.0%)
[*] Waiting for new xApp connection...
Capturing on '5g-oran'
```

② Start xApp

Once all RAN components are running, we update the RIC endpoint IP address so that the xApp communicates with the MiTM proxy. Specifically, we modify line **53** of the *docker-compose.yaml* file, as shown in Listing 8.

Listing 8: *docker-compose.yaml* content example

```
xapp-kpm-monitor:
  profiles: [xapp]
  image: researchanon2025/flexric-components:v1.0
  restart: unless-stopped
  tty: true
  privileged: true
  command: /bin/bash -c
          "./build/examples/xApp/c/monitor/xapp_kpm_moni
- -c ./configs/flexric.conf
+ -c ./configs/xapp.conf # overrides RIC IP to localhost
          && sleep infinity"
  security_opt:
    - seccomp:unconfined
  volumes:
    - ./configs:/flexric/configs
  networks:
    core_net:
      ipv4_address: 192.168.71.185
```

Finally, start an xApp KPI monitoring example by running the command below:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
docker compose --profile xapp up # Terminal 4 - xAPP
```

The xApp informs the downlink and uplink throughput of all connected UEs as shown below:

```
xapp-kpm-monitor-1  | ran_ue_id = 1
xapp-kpm-monitor-1  | DRB.RlcSduDelayDl = 6014.82 [μs]
xapp-kpm-monitor-1  | DRB.UEThpDl = 250493.20 [kbps]
xapp-kpm-monitor-1  | DRB.UEThpUl = 4843.06 [kbps]
xapp-kpm-monitor-1  | RRU.PrbTotDl = 354196 [PRBs]
xapp-kpm-monitor-1  | RRU.PrbTotUl = 30572 [PRBs]
```

③ Automated testing

The previous steps manually set up all required containers and start the SCTP MiTM proxy within a single session. Note the use of *sleep infinity* in the `xapp-kpm-monitor` container, which keeps the xApp running indefinitely. To simplify this process, we provide a short script that automates both the Docker setup and the fuzzing sessions. To run it, simply execute:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
python3 docker_monitoring.py
```

This script launches an infinite fuzzing session, periodically restarting the xApp every 15 seconds to create multiple independent sessions. All outputs produced during fuzzing are stored in well-organized directories: packet captures are saved in ***captures_bridge***, Docker logs from the OAI containers are stored in ***container_logs***, and fuzzer logs are recorded in ***fuzzing_logs***.
The learned E2 protocol state machines are saved in ***state_machines***, while behavioral differences between normal and fuzzed sessions are stored in ***state_machines_diff*** for further analysis.
The output should resemble the following:

Listing 9: Example of *docker_monitoring.py* output

```
2026-01-25 15:51:37,686 - INFO -
=== 5G O-RAN Synchronized Monitor Started ===
2026-01-25 15:51:37,686 - INFO - Container logs: ././asn1/container_logs/
```

```
2026-01-25 15:51:37,686 - INFO - Fuzzing logs: ././asn1/fuzzing_logs/
2026-01-25 15:51:37,693 - INFO -
Starting synchronized monitor...
2026-01-25 15:51:37,693 - INFO - Auto-recovery: oai-du-rfsim, oai-cu-cp, oai-cu-up, nearRT-RIC
2026-01-25 15:51:37,695 - INFO - Starting services...
2026-01-25 15:51:37,695 - WARNING -  Missing: ['mongo', 'nrf', 'scp', 'upf', 'smf', 'pcrf', 'amf', 'ausf\
    ', 'udm', 'pcf', 'nssf', 'bsf', 'udr', 'webui', 'oai-du-rfsim', 'oai-cu-cp', 'oai-cu-up', '\
    oai-ue-rfsimu-1', 'oai-ue-rfsimu-2', 'nearRT-RIC', 'xapp-kmp-monitor']
2026-01-25 15:51:37,695 - INFO - Executing: docker compose --profile core up -d
WARN[0000] No services to build
WARN[0000] Found orphan containers ([proxy oranclaw-e2-mitm-fuzzing-fuzzer-ui-1 o-ran-fuzzer]) for this \
    project. If you removed or renamed this service in your compose file, you can run this command with\
     the --remove-orphans flag to clean it up.
[+] up 15/15
 Network 5g-oran-net                        Created                                                      \
                   0.0s
 Container oranclaw-e2-mitm-fuzzing-mongo-1 Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-webui-1 Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-nrf-1   Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-scp-1   Created                                                      \
                   0.0s
 Container oranclaw-e2-mitm-fuzzing-upf-1   Healthy                                                      \
                   2.7s
 Container oranclaw-e2-mitm-fuzzing-smf-1   Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-amf-1   Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-pcrf-1  Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-ausf-1  Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-udm-1   Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-pcf-1   Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-nssf-1  Created                                                      \
                   0.1s
 Container oranclaw-e2-mitm-fuzzing-bsf-1   Created                                                      \
                   0.0s
 Container oranclaw-e2-mitm-fuzzing-udr-1   Created                                                      \
                   0.0s
2026-01-25 15:51:42,140 - INFO - Success: docker compose --profile core up -d
2026-01-25 15:51:45,141 - INFO - Executing: docker compose --profile gnb-rfsim up -d &
2026-01-25 15:51:45,142 - INFO - Success: docker compose --profile gnb-rfsim up -d &
WARN[0000] No services to build
WARN[0000] Found orphan containers ([proxy oranclaw-e2-mitm-fuzzing-fuzzer-ui-1 o-ran-fuzzer]) for this \
    project. If you removed or renamed this service in your compose file, you can run this command with\
     the --remove-orphans flag to clean it up.
[+] up 6/7
[+] up 7/7 ue-rfsim-net                                 Created                                          \
                    0.0s
 Network ue-rfsim-net                                Created                                             \
                  0.0s
 Container oranclaw-e2-mitm-fuzzing-nearRT-RIC-1     Created                                             \
                  0.1s
 Container oranclaw-e2-mitm-fuzzing-oai-cu-cp-1      Healthy                                             \
                  2.9s
 Container oranclaw-e2-mitm-fuzzing-oai-du-rfsim-1   Created                                             \
                  0.1s
[+] up 6/7er oranclaw-e2-mitm-fuzzing-oai-cu-up-1       Created                                          \
                    0.1s
[+] up 7/7 ue-rfsim-net                                 Created                                          \
                    0.0s
 Network ue-rfsim-net                                Created                                             \
                  0.0s
 Container oranclaw-e2-mitm-fuzzing-nearRT-RIC-1     Created                                             \
                  0.1s
 Container oranclaw-e2-mitm-fuzzing-oai-cu-cp-1      Healthy                                             \
```

```
                  2.9s  Container oranclaw-e2-mitm-fuzzing-oai-du-rfsim-1    Healthy                    \
                                          5.6s
 Container oranclaw-e2-mitm-fuzzing-oai-cu-up-1       Created                                          \
                  0.1s
 Container oranclaw-e2-mitm-fuzzing-oai-ue-rfsimu-1-1 Created                                          \
                  0.1s
 Container oranclaw-e2-mitm-fuzzing-oai-ue-rfsimu-2-1 Created                                          \
                  0.1s
2026-01-25 15:51:51,541 - INFO - Started log process for oranclaw-e2-mitm-fuzzing-oai-cu-up-1 -> ./asn1/\
    container_logs/oranclaw-e2-mitm-fuzzing-oai-cu-up-1_20260125_155151.log
2026-01-25 15:51:51,541 - INFO - Started log process for oranclaw-e2-mitm-fuzzing-oai-du-rfsim-1 -> ./\
    asn1/container_logs/oranclaw-e2-mitm-fuzzing-oai-du-rfsim-1_20260125_155151.log
2026-01-25 15:51:51,542 - INFO - Started log process for oranclaw-e2-mitm-fuzzing-oai-cu-cp-1 -> ./asn1/\
    container_logs/oranclaw-e2-mitm-fuzzing-oai-cu-cp-1_20260125_155151.log
2026-01-25 15:51:51,542 - INFO - Started log process for oranclaw-e2-mitm-fuzzing-nearRT-RIC-1 -> ./asn1\
    /container_logs/oranclaw-e2-mitm-fuzzing-nearRT-RIC-1_20260125_155151.log
2026-01-25 15:51:51,542 - INFO -  Starting MiTM fuzzing session: 20260125_155151
2026-01-25 15:51:51,543 - INFO -  MiTM session started
2026-01-25 15:51:51,568 - INFO -
Container                                  Status         Health
2026-01-25 15:51:51,568 - INFO - ----------------------------------------------------------------------
2026-01-25 15:51:51,568 - INFO - oranclaw-e2-mitm-fuzzing-xapp-kpm-monitor-1running        N/A
2026-01-25 15:51:51,568 - INFO - oranclaw-e2-mitm-fuzzing-oai-ue-rfsimu-2-1running        starting
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-oai-ue-rfsimu-1-1running        starting
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-oai-cu-up-1    running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-oai-du-rfsim-1 running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-oai-cu-cp-1    running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-nearRT-RIC-1   running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-udr-1          running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-bsf-1          running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-nssf-1         running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-pcf-1          running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-udm-1          running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-ausf-1         running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-amf-1          running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-pcrf-1         running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-smf-1          running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-upf-1          running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-scp-1          running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-webui-1        running        N/A
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-nrf-1          running        healthy
2026-01-25 15:51:51,569 - INFO - oranclaw-e2-mitm-fuzzing-mongo-1        running        healthy
2026-01-25 15:51:51,569 - INFO - ----------------------------------------------------------------------
2026-01-25 15:51:51,569 - INFO - MiTM Session: 20260125_155151
2026-01-25 15:51:51,569 - INFO - ----------------------------------------------------------------------
```

Furthermore, the fuzzer begins saving log files in the ***fuzzing_logs*** directory (e.g., *mitm_session_YYYYMMDD_HHMMSS.log*). These logs show that, once the fuzzing process starts, the fuzzer actively analyzes network traffic and receives, mutates, and forwards messages. An example of this output is shown in Listing 10.

Listing 10: Example fuzzing log output

```
==================================================
            SCTP MITM Proxy - Starting
==================================================
[+] Started tshark capture
[Genetic Algorithm] Enabled
[+] Connected to RIC at 192.168.71.184:36422
[*] Listening for xApp on 192.168.71.129:36422...
[*] Fuzzing enabled: True (Probability: 20.0%)
[*] Waiting for new xApp connection...
Capturing on '5g-oran'
[+] Accepted xApp connection from ('192.168.71.185', 54189)
[TX] [xApp --> RIC]
[DEBUG]Length: 492 bytes
[*] ProcedureCode: 0xe
[DEBUG] Message Type: E42setup
--------------------------------------------------
[RX] [xApp <-- RIC]
```

```
[DEBUG]Length: 2389 bytes
[*] ProcedureCode: 0xe
[DEBUG] Response Type: E42setup
[DEBUG] Overhead: 0.32 ms
--------------------------------------------------
[TX] [xApp --> RIC]
[DEBUG]Length: 134 bytes
[*] ProcedureCode: 0xf
[DEBUG] Message Type: E42RICsubscription
[*] ProcedureCode: 0xf
[*] Processing E42RICsubscription message (134 bytes)
Before decoding:
000f008081000005003e000200070003000c0499f9075000000e00200e00001d000500000100000005000200020001e0055000308\
    03e7000013004b4000004700010480420000003840010004010100000101204452422e50646370536475566f6c756d65444\
    c012000000120445242 2e50646370536475566f6c756d65554c012000004003e7
Converted to JSON:
{
  "initiatingMessage": {
    "procedureCode": 15,
    "criticality": "reject",
    "value": {
      "protocolIEs": [
        {
          "id": 62,
          "criticality": "reject",
          "value": 7
        },
        {
          "id": 3,
          "criticality": "reject",
          "value": {
            "gNB": {
              "global-gNB-ID": {
                "plmn-id": "99f907",
                "gnb-id": {
                  "gnb-ID": {
                    "value": "00000e00",
                    "length": 32
                  }
                }
              },
              "gNB-CU-UP-ID": 3584
            }
          }
        },
        {
          "id": 29,
          "criticality": "reject",
          "value": {
            "ricRequestorID": 1,
            "ricInstanceID": 0
          }
        },
        {
          "id": 5,
          "criticality": "reject",
          "value": 2
        },
        {
          "id": 30,
          "criticality": "reject",
          "value": {
            "ricEventTriggerDefinition": "0803e7",
            "ricAction-ToBeSetup-List": [
              {
                "id": 19,
                "criticality": "reject",
                "value": {
                  "ricActionID": 0,
                  "ricActionType": "report",
                  "ricActionDefinition": "000104804200000038400100040101000001012044524 22e50646370536475\
```

```
    566f6c756d65444c012000000120445242 2e50646370536475566f6c756d65554c012000004003e7"
                }
              }
            ]
          }
        }
      ]
    }
  }
}
[*] Selected fuzzing strategy: asn1
[+] Mutated field 'granulPeriod' to 2012211191 in E2SM-KPM-ActionDefinition
[+] Encoded again:
000f008083000005003e000200070003000c0499f9075000000e00200e00001d000500000100000005000200020 01e0057000308\
    03e7000013004d4000004900010480440000003840010004010100000101204452422e50646370536475566f6c756d65444\
    c012000000120445242 2e50646370536475566f6c756d65554c01200000c077efe7f6
--------------------------------------------------
```

2.3. Evaluating *ORANClaw* with ns-3 O-RAN Simulator

① Install ns-3 O-RAN Simulator

Building from source is required to evaluate the ns-3 O-RAN simulator. We evaluated *ORANClaw* against the Orange OpenSource ns-O-RAN implementation at commit `7936f62f`, which integrates ns-3 with the FlexRIC near-RT RIC.

```
cd $HOME
git clone https://github.com/Orange-OpenSource/ns-O-RAN-flexric.git
cd ns-O-RAN-flexric
git checkout 7936f62f
```

Detailed build and installation instructions are provided in the official repository documentation.

② Configure Fuzzing Engine for ns-3

Once ns-3 is successfully built and running, the same *ORANClaw* fuzzing methodology described in Section 2.2 can be applied. The only required change is to configure the fuzzing engine to target the ns-3 E2 termination endpoint instead of the OAI base station. Since both FlexRIC and the ns-3 gNB/eNB listen on the local host interface (`127.0.0.1`), the MITM component must be configured to use the host machine's IP address.
Specifically, modify line *21* of the `client_server_ns3.py` script as follows:

Listing 11: `client_server_ns3.py` configuration

```
IP_MITM = "192.168.71.129"  # change to your host IP
IP_XAPP = "127.0.0.1"
IP_RIC  = "127.0.0.1"
```

③ Update ASN.1 Specifications for ns-3 Compatibility

```
ORANClaw-E2-MitM-Fuzzing ................Artifact repository for ORANClaw E2 MitM fuzzing framework
└─ asn1 ..............................................ASN.1 specifications, code generation, and fuzzing logic
   └─ asn1files ......................................................E2AP and E2SM ASN.1 specifications
      ├─ e2ap-v01.01.asn
      └─ e2sm_kpm_v03.00.asn
```

Figure 2: ASN.1 files

Regarding FlexRIC, once reaching Section 1.3 of its installation guide, note that FlexRIC is configured by default to build the near-RT RIC using *E2AP v2.03* and *E2SM-KPM v2.03*. In contrast, the ns-O-RAN simulator relies on *E2AP v1.01* and *E2SM-KPM v3.00*. To ensure compatibility, *ORANClaw* must regenerate its encoder and decoder using the corresponding ASN.1 specifications.
This can be achieved by modifying the ASN.1 input files referenced in the `gen-code.sh` script. The relevant files are located under the `asn1files` directory, shown in Figure 2.
Concretely, the existing lines in `gen-code.sh` can be commented out and replaced by the following commands, inserted before line *46* of the script:

```
./asn1c/bin/asn1c asn1files/e2ap-v01.01.asn asn1files/e2sm_kpm_v03.00.asn -genTest \
 -genTest \
-per -json -w32 -server -depends -events -genPrtToStr \
-list -reader -trace -c++ -print -prtfmt details \
-srcdir src -make Makefile -w32 -table-unions -stream -pdu all
```

After updating the script, regenerate and build the encoder and decoder as follows:

```
cd $HOME/ORANClaw-E2-MitM-Fuzzing-AFFC/
./gen-code.sh
```

This process generates the appropriate encoding and decoding logic required for *ORANClaw* to correctly process E2 messages exchanged with the ns-3 O-RAN simulator.

2.3.1. Changing the baseline capture. Since *ORANClaw* needs a baseline state machine to generate the diffs in the `SM_DIFF` module, this trace is different from OAI implementation. Its a different FlecRIC xApp, speficied in ns3-ORAN repository. More precisely, xApp using RC service model.

To this end, simply modify line 1736 of the script *client_server_ns3.py*. This will select another *pcapng* file to make the diffs on the `SM_DIFF` component of *ORANClaw*.

Listing 12: `wdmapper_gen_diff() function update example`

```
        wdmapper_gen_diff = [
            "./bin/wdmapper",
            "--udp-dst-port=36421,38412,9999,38472,38412,36422",
            "-d", current_file,
- "-i", '../../../captures/Baseline_kpm_mon.pcapng',
+ "-i", '../../../captures/Baseline_kpm_rc_ns3.pcapng',
            "-c", "./configs/5gnr\_gnb\_config.json",
            "-o", (os.path.join(self.state\_machines\_diff\_folder,f"wdmapper\_diff\_{self.timestamp}.\
    svg"))
        ]
```

NOTE: The evaluation machine has already all ns3 libraries and *ORANClaw* code pre-installed under repository *oran-orachestration/*. There is no need to re-compile and build the E2AP and E2SM on this folder.

4 Run Automated Fuzzing

```
cd $HOME/oran-orchestration/asn1
./ns3_fuzzing.sh
```

This script will do a similar task as when we evaluated OAI. Restarting the xApp every 15 seconds, saving the logs in the same correspondent files for further analysis.

Listing 13: Example of `client_server_ns3.py` script output

```
==================================================
            SCTP MITM Proxy - Starting
==================================================
[+] Created capture files folder: ns3/oran-orchestration/asn1/captures_bridge
[+] Created state machines folder: ns3/oran-orchestration/asn1/state_machines
[+] Created state machines diff folder: ns3/oran-orchestration/asn1/state_machines_diff
[+] Started tshark capture
[+] Connected to RIC at 127.0.0.1:36422
[*] Listening for xApp on 192.168.1.155:36422...
[*] Fuzzing enabled: True (Probability: 20.0%)
[*] Waiting for new xApp connection...
Capturing on 'Loopback: lo'
[+] Accepted xApp connection from ('192.168.1.155', 40907)
[TX] [xApp --> RIC]
[DEBUG][39mLength: 509 bytes
[*] ProcedureCode: 0xc
[DEBUG] Message Type: E42setup
[*] ProcedureCode: 0xc
```

```
[*] Processing E42setup message (509 bytes)
Before decoding:
000c0081f8000001000a0081f000080008005c8000023800304f52414e2d4532534d2d4b504d000018312e332e362e312e342e31\
    2e35333134382e312e332e322e3205004b504d204d6f6e69746f720003011c000018312e332e362e312e342e312e3533313\
    4382e312e332e322e320008005c800003380005804f52414e2d4532534d2d5243000018312e332e362e312e342e312e3533\
    3134382e312e312e322e33050052414e20436f6e74726f6c0001011c000018312e332e362e312e342e312e35333134382e3\
    12e312e322e330008002e80008e0c4d41435f53544154535f56300001011a000016302e302e302e302e302e302e302e\
    312e3134322e300008002e80008f0c524c435f53544154535f56300001011a000016302e302e302e302e302e302e302\
    e312e3134332e300008002f8000900d504443505f53544154535f56300001011a000016302e302e302e302e302e302e\
    302e312e3134342e30000800308000910e534c4943455f53544154535f56300001011a000016302e302e302e302e302\
    e302e302e312e3134352e300008002d8000920b54435f53544154535f56300001011a000016302e302e302e302e302e\
    302e302e312e3134362e300008002e8000940c4754505f53544154535f56300001011a000016302e302e302e302e302\
    e302e302e312e3134382e30
Converted to JSON:
{
  "initiatingMessage": {
    "procedureCode": 12,
    "criticality": "reject",
    "value": {
      "protocolIEs": [
        {
          "id": 10,
          "criticality": "reject",
          "value": [
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 2,
                "ranFunctionDefinition": "00304f52414e2d4532534d2d4b504d000018312e332e362e312e342e312e35\
    333134382e312e332e322e3205004b504d204d6f6e69746f72",
                "ranFunctionRevision": 3,
                "ranFunctionOID": "1.3.6.1.4.1.53148.1.3.2.2"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 3,
                "ranFunctionDefinition": "0005804f52414e2d4532534d2d5243000018312e332e362e312e342e312e35\
    333134382e312e312e322e33050052414e20436f6e74726f6c",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "1.3.6.1.4.1.53148.1.1.2.3"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 142,
                "ranFunctionDefinition": "4d41435f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.142.0"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 143,
                "ranFunctionDefinition": "524c435f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.143.0"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 144,
```

```
                "ranFunctionDefinition": "504443505f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.144.0"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 145,
                "ranFunctionDefinition": "534c4943455f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.145.0"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 146,
                "ranFunctionDefinition": "54435f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.146.0"
              }
            },
            {
              "id": 8,
              "criticality": "reject",
              "value": {
                "ranFunctionID": 148,
                "ranFunctionDefinition": "4754505f53544154535f5630",
                "ranFunctionRevision": 1,
                "ranFunctionOID": "0.0.0.0.0.0.0.0.1.148.0"
              }
            }
          ]
        }
      ]
    }
  }
}
```


```
[*] Randomized weights for fuzzing strategies: [0.53, 0.38, 0.09]
[*] Selected fuzzing strategy: json
[DEBUG] Selected JSON fields to mutate: ['criticality']
[DEBUG] Generated new value for criticality: notify
[+] Mutated initiatingMessage.value.protocolIEs[0].value[0].criticality: reject -> notify
[+] Successfully fuzzed message (509 bytes)
[+] Encoded again:
```


```
000c0081f8000001000a0081f000080008805c8000023800304f52414e2d4532534d2d4b504d000018312e332e362e312e342e31\
    2e35333134382e312e332e322e3205004b504d204d6f6e69746f720003011c000018312e332e362e312e342e312e3533313\
    4382e312e332e322e320008005c800003380005804f52414e2d4532534d2d5243000018312e332e362e312e342e312e3533\
    3134382e312e312e322e33050052414e20436f6e74726f6c0001011c000018312e332e362e312e342e312e35333134382e3\
    12e312e322e330008002e80008e0c4d41435f53544154535f56300001011a000016302e302e302e302e302e302e302e302e\
    312e3134322e300008002e80008f0c524c435f53544154535f56300001011a000016302e302e302e302e302e302e302e302\
    e312e3134332e300008002f8000900d504443505f53544154535f56300001011a000016302e302e302e302e302e302e302e\
    302e312e3134342e30000800308000910e534c4943455f53544154535f56300001011a000016302e302e302e302e302e302\
    e302e302e312e3134352e300008002d8000920b54435f53544154535f56300001011a000016302e302e302e302e302e302e\
    302e302e312e3134362e300008002e8000940c4754505f53544154535f56300001011a000016302e302e302e302e302e302\
    e302e302e312e3134382e30
```


```
[DEBUG] Overhead: 6.74 ms
```

3. Evaluation of effectiveness in finding bugs (RQ1)

Our objective is to replicate the effectiveness of the fuzzing sessions, as presented in Figures 4 and 5 in the paper. In this section, we provide details of the steps we followed to obtain the data in. In summary, we collected the logs for each implementation (OAI and ns3) under ***Logs_ORANCLAW*** directory.
To facilitate replication of our results, all the logs used in the experiments can be downloaded from the following location:

```
cd $HOME
# The zip file is around 38.3 GB
wget -O Logs_ORANCLAW.zip https://zenodo.org/records/18389642/files/Logs_ORANCLAW.zip?download=1
unzip -p Logs_ORANCLAW.zip | tar -xvf -
```

Then, uncompress the data (492.4 GB).
We note that, to generate the figures in the paper, the provided scripts are designed to scan all log files to identify crashes, assertions, and related events. As a result, each script takes a considerable amount of time to complete.

3.1. Log Organization

Figure 3 shows a simplified view of the log directory structure, highlighting only the directories that contain Python scripts used for aggregating results and generating the figures reported in RQ2.
The directory structure mirrors that used for RQs, with separate subdirectories for OpenAirInterface and ns-3. Each ablation variant contains multiple runs (e.g., *Rand1*, *Rand5*)

```
Logs_ORANCLAW ..............................................Pre-collected logs and raw data
├── Logs_NS3 ...............................................ns-3 ablation experiments
│   ├── Ablation_2 .........................................Extended ablation study
│   │   └── plot.py ........................................ns3 Ablation, Figure 7 of the paper
│   └── RQ1 ................................................RQ1/RQ2 shared analysis utilities
│       └── average.py .....................................ns3 cummulative bugs, Figure 5 of the paper
└── LogsOAI ................................................OAI ablation experiments
    ├── Ablation ...........................................OAI ablation analysis
    │   ├── plot.py ........................................Ablation result visualization (Figure 6 of the paper)
    │   └── sm_completness.py ..............................State Machine completness (Figure 8 of the paper)
    ├── RQ1 ................................................OAI statistical analysis
    │   └── average_new_SM.py ..............................OAI cummulative bugs, Figure 4 of the paper
    └── RQ3 ................................................Efficiency of the bridge
        └── latency_wisec.py ...............................Efficiency boxplot (Figure 9 of the paper)
```

Figure 3: Simplified organization of *ORANClaw* log directories showing only Python scripts used for analysis and figure generation in the ablation study (RQ2).

NOTE: Due to the inherent stochastic behavior of the fuzzer, the exact number and timing of crashes may vary between runs. Nevertheless, all the relevant logs used in our experiments are provided, allowing readers to reproduce the plots and verify the trends reported in the paper.

3.2. OpenAirInterface

You can create your fuzzing sessions and trigger the vulnerabilities in real time by using the script in section 2.2. To facilitate replication of our results, we provide an automation script located at *Logs_ORANCLAW/LogsOAI/RQ1/average_new_SM.py* that generates Figure 4 of the paper. Users can simply run this script after ensuring that the log directories are organized according to the relative paths expected by the script. The script automatically determines the base path relative to its own location, as depicted in Listing 14.

Listing 14: *average_new_SM.py* script content example

```
# Automatically detect base path relative to script
SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))
DEFAULT_PATH = os.path.abspath(os.path.join(SCRIPT_DIR, '..'))
BASE_PATH = os.getenv('LOG_BASE_PATH', DEFAULT_PATH)

SCENARIOS = {
    "Mutation": {
```

```
        "runs": [
            os.path.join(BASE_PATH, "RQ1/OAIRand1/container_logs/"),
            os.path.join(BASE_PATH, "RQ1/OAIRand2/container_logs/"),
            os.path.join(BASE_PATH, "RQ1/OAIRand3/container_logs/"),
            os.path.join(BASE_PATH, "RQ1/OAIRand4/container_logs/"),
            os.path.join(BASE_PATH, "RQ1/OAIRand5/container_logs/"),
        ],
        "color": "orange"
    },
    # ... other scenarios
}
```

If users prefer to store logs in a different location, they can override the base path by setting the `LOG_BASE_PATH` environment variable before running the script. For example, if logs are stored at */custom/path/to/Logs_ORANCLAW/LogsOAI*:

```
source $HOME/.venvs/oran-env/bin/activate
export LOG_BASE_PATH=/custom/path/to/Logs_ORANCLAW/LogsOAI
python3 average_new_SM.py
```

Alternatively, if using the default structure where the script is in *Logs_ORANCLAW/LogsOAI/RQ1/*, you can run it directly:

```
source $HOME/.venvs/oran-env/bin/activate
cd $HOME/Logs_ORANCLAW/LogsOAI/RQ1
python3 average_new_SM.py
```

The script will automatically detect that the base path is one directory level up (i.e., *Logs_ORANCLAW/LogsOAI/*) and locate all required log directories from there.
Listing 16 ilustrates the output of the script. Where it will process the logs under *container_logs* directory.

Listing 15: *average_new_SM.py* output example

```
Using BASE_PATH: /Logs_ORANCLAW/LogsOAI
Processing Mutation scenario...
  Processing run 1/5: /Logs_ORANCLAW/LogsOAI/RQ1/OAIRand1/container_logs/
  Time range: 2025-07-29 09:33:57 to 2025-07-30 09:33:28 (24.0h)
  Found 31 unique crashes, 31 within 24.0h window
  Processing run 2/5: /Logs_ORANCLAW/LogsOAI/RQ1/OAIRand2/container_logs/
  Time range: 2025-07-29 09:34:01 to 2025-07-30 09:43:13 (24.2h)
  Found 24 unique crashes, 24 within 24.0h window
  Processing run 3/5: /Logs_ORANCLAW/LogsOAI/RQ1/OAIRand3/container_logs/
  Time range: 2025-07-30 09:55:59 to 2025-07-31 09:55:57 (24.0h)
  ...
```

Thius results in the statistics that correspond to Figure 4 of the paper and reflect the crash discovery behavior observed during the evaluation in OAI implementation.

3.3. ns-3

Users can create their own fuzzing sessions for the ns-3 platform by following the instructions in Section 2.3. As with OpenAirInterface (Section 3.2), we provide an automation script to process the logs generated during the fuzzing campaigns and to identify crashes and anomalies in a consistent manner.
Specifically, we include a Python script that aggregates results across multiple ns-3 fuzzing runs and computes the average number of unique crashes over time, corresponding to Figure 5 in the paper. Once the log directories are organized according to the expected structure, the script can be executed as follows:

```
source $HOME/.venvs/oran-env/bin/activate
cd $HOME/Logs_ORANCLAW/Logs_NS3/RQ1/
python3 average.py
```

Similar to the OpenAirInterface setup, the script automatically determines the base log directory and iterates over multiple independent fuzzing runs (e.g., `Ns3Opt1`–`Ns3Opt5`). For each run, it extracts timestamps, normalizes execution time to a 24-hour window, and counts the number of unique crashes observed within that interval.
Listing 16 shows an example of the script output. The script processes logs under the `container_logs` directories, reports the duration of each fuzzing run, and computes the average number of crashes across valid runs:

Listing 16: *average.py* output example

```
Using BASE_PATH: Logs_ORANCLAW/Logs_NS3
```

```
Processing Mutation scenario...
  Processing run 1/5: Logs_ORANCLAW/Logs_NS3/RQ1/Ns3Opt1/container_logs/
  Time range: 2025-07-25 14:57:48 to 2025-07-26 15:00:33 (24.0h)
  Found 31 unique crashes, 31 within 24.0h window
  Processing run 2/5: Logs_ORANCLAW/Logs_NS3/RQ1/Ns3Opt2/container_logs/
  Time range: 2025-07-25 14:58:08 to 2025-07-26 15:00:37 (24.0h)
  Found 29 unique crashes, 29 within 24.0h window
  ...
  Using 5/5 valid runs
  Normalizing to 24.0 hours (shortest run duration)
  Completed Mutation: Average final count = 31.8
```

The output of the script, corresponds to the ns-3 evaluation results reported in the paper, allowing readers to reproduce Figure 5 and verify the effectiveness of *ORANClaw* in discovering implementation bugs in a ns3 simulator O-RAN environment.

4. Evaluation the effectiveness of *ORANClaw* components (RQ2)

This section describes how to reproduce the ablation study presented in RQ2, which evaluates the individual contribution of *ORANClaw* 's fuzzing components. In particular, we study the impact of mutation, duplication, and optimization strategies, as well as the effect of modifying the parameters of the cost function and the genetic optimizer. All configurations are implemented directly in the fuzzing engine and can be adjusted to enable or disable specific components.

Listing 17: *calculate_fuzzing_cost* function in `client_server.py` (line 1867)

```
def calculate_fuzzing_cost(self, target_fields=None, max_mutations=1,
                           alpha=0, beta=40, mu=0.1, t=20, timestamp=None):
    """
    Calculate the cost/reward function for fuzzing based on state machine exploration.
    """
    ...
```

The cost function used to guide fuzzing is implemented in the *calculate_fuzzing_cost* function in *client_server.py* (line 1867). The parameters of this function such as α, β, μ, and τ control the relative importance of state changes, protocol transitions, and mutation complexity during fuzzing. By modifying these parameters, users can directly reproduce the different optimization variants evaluated in RQ2, including configurations where transition-based guidance ($\beta = 0$) or mutation penalties ($\mu = 0$) are disabled.

Listing 18: Genetic optimizer configuration in `client_server.py` (line 128)

```
class GeneticFuzzerOptimizer:
    def __init__(self, population_size=6,
                 mutation_rate=0.3):
        ...
```

In addition, the parameters of the genetic optimization component can be customized in the *GeneticFuzzerOptimizer* class (line 128 in *client_server.py*). Parameters such as the population size and mutation rate allow users to control the behavior of the evolutionary search process and were adjusted to evaluate the contribution of optimization in the ablation study.

For each configuration, we ran multiple independent fuzzing sessions under the same time experimental setup, varying only the enabled fuzzing components (mutation, duplication, and optimization) and the associated optimization parameters (e.g., β, μ). All logs are organized under the *Logs_ORANCLAW* directory using a consistent directory structure.

To reproduce the ablation results reported in RQ2 (Figures 6, 7, and 8), we provide pre-collected logs together with automation scripts that aggregate crash statistics for the different *ORANClaw* variants.

4.1. OpenAirInterface

To reproduce the ablation results for OpenAirInterface (corresponding to Figure 6 of the paper), users can run the provided script located in:

```
cd $HOME/Logs_ORANCLAW/LogsOAI/Ablation/
source $HOME/.venvs/oran-env/bin/activate
python3 plot.py
```

The script automatically detects the base log directory, iterates over all ablation variants and runs, normalizes execution time to a 24-hour window, and computes the average number of unique crashes over time.

5. ns3

Similarly, reproduce the ns-3 ablation study (Figure 7) using:

```
cd $HOME/Logs_ORANCLAW/Logs_NS3/Ablation_2/
source $HOME/.venvs/oran-env/bin/activate
python3 plot.py
```

Listing 19 shows an example output produced by the ns-3 ablation aggregation script, illustrating how unique crashes are identified and saved for each fuzzing configuration.

Listing 19: Example output of ns-3 ablation aggregation script

```
Using BASE_PATH: /media/p3rplex/data/Logs_ORANCLAW/Logs_NS3
Processing Optimization...
Optimization: 34 unique crashes saved to unique_crashes_opt.csv
Processing Mutation...
Mutation: 32 unique crashes saved to unique_crashes_random.csv
Processing Optimization + Dup...
Optimization + Dup: 34 unique crashes saved to unique_crashes_opt_mut_dup.csv
Processing Mutation + Dup...
Mutation + Dup: 33 unique crashes saved to unique_crashes_rand_mut_dup.csv
Processing Dup Only...
```

5.1. State-machine completeness analysis (OAI)

To reproduce the state-machine completeness study (Figure 8 of the paper), run the dedicated script that compares fuzzing runs generated from complete and incomplete state-machine traces. The script aggregates results across trace configurations and produces the corresponding crash discovery curves:

```
cd $HOME/Logs_ORANCLAW/LogsOAI/Ablation/
source $HOME/.venvs/oran-env/bin/activate
python3 sm_completness.py
```

The script processes all relevant ablation variants, including optimization-only, mutation-only, duplication-only, combined configurations, and variants with transition-based guidance disabled (e.g., $\beta = 0$). The resulting plots quantify the impact of state-machine completeness on crash discovery and correspond to RQ2.
Example output (state-machine completeness).. Listing 20 shows an example output of the state-machine completeness analysis script, indicating the processing of different fuzzing configurations including the $\beta = 0$ variant.

Listing 20: Example output of state-machine completeness analysis

```
Using BASE_PATH: /media/p3rplex/data/Logs_ORANCLAW/LogsOAI
Processing Optimization...
Processing Mutation...
Processing Dup Only...
Processing Mutation + Dup...
Processing Optimization + Dup...
Processing Optimization ($\beta$=0)...
```

6. Evaluation of efficiency of *ORANClaw* (RQ3)

This section describes how to reproduce the efficiency evaluation of *ORANClaw* presented in RQ3. The goal of this experiment is to quantify the per-packet latency overhead introduced by *ORANClaw* when operating as an inline E2 bridge between the xApp and the RIC. While absolute latency values may vary depending on hardware and system load, the overhead measurements extracted from the logs consistently show that *ORANClaw* introduces only tens of milliseconds of additional latency when performing full packet processing. This overhead remains within the timing requirements of the E2 interface, confirming that *ORANClaw* can be safely deployed on time-critical RAN control paths.

6.1. Latency Measurement Methodology

ORANClaw operates as a man-in-the-middle SCTP proxy on the E2 interface. For every E2AP packet forwarded through the bridge, *ORANClaw* logs the processing latency incurred between packet reception and packet transmission. This latency is reported directly in the runtime logs as a per-packet metric.
Specifically, for each transmitted or received packet, the bridge prints a log entry of the form:

```
[DEBUG] Overhead: X ms
```

where `X` represents the elapsed time (in milliseconds) spent inside the bridge. This duration includes socket reception, optional ASN.1 decoding, fuzzing logic execution, re-encoding, and packet forwarding.
Latency is recorded for both transmission directions:

- **TX**: packets sent from the RIC to the gNB, and
- **RX**: packets received from the gNB and forwarded to the RIC.

6.2. Fuzzzed vs Unproccessed

Two execution modes are evaluated:

1) **Unproccessed messages**, where packets are forwarded without decoding or mutation. In this mode, the logs indicate normal forwarding (e.g., Sent packet normally), and the reported overhead reflects only minimal proxy handling.
2) **Fuzzed messages**, where fuzzing is enabled and packets undergo full ASN.1 decoding, JSON conversion, mutation, and re-encoding before forwarding. In this case, the logs explicitly show decoding steps, selected fuzzing strategies, mutated fields, and a corresponding increase in reported overhead.

An excerpt of the runtime logs illustrates this distinction. For example, simple forwarding incurs sub-millisecond overhead:

```
[DEBUG] Overhead: 0.50 ms
```

whereas packets undergoing full processing exhibit higher latency:

```
[DEBUG] Overhead: 22.67 ms
```

These per-packet measurements form the basis of the latency analysis in RQ3.

6.3. Log Collection

Latency logs are collected continuously during long-running fuzzing sessions (24 hours in our evaluation). Each session may include multiple xApp connections, reconnections, and thousands of E2AP packets. The overhead values are emitted inline with other diagnostic messages, such as procedure codes, message types, and fuzzing actions.
All raw logs used in this evaluation are pre-collected and included in the artifact under:

```
Logs_ORANCLAW/LogsOAI/RQ3/
```

6.4. Latency Analysis and Plot Generation

To generate the latency boxplot shown in Figure 9 of the paper, we provide a Python script that parses the runtime logs and extracts all reported overhead values. The script identifies lines containing the `[DEBUG] Overhead` tag and aggregates the corresponding latency measurements across the entire experiment.
Users can reproduce the plot by executing:

```
cd $HOME/Logs_ORANCLAW/LogsOAI/RQ3/
source $HOME/.venvs/oran-env/bin/activate
python3 latency_wisec.py
```

The script separates latency samples corresponding to bypass and processing modes, and generates a combined boxplot comparing the two configurations. All measurements are retained, including extreme values, to accurately reflect worst-case behavior during sustained fuzzing.
The resulting figure corresponds directly to Figure 9 in the paper.

ORANClaw: Shredding E2 Nodes in O-RAN via Structure-aware MiTM Fuzzing

Anonymous

Abstract

The open radio access network (O-RAN) standard provides a foundational move towards disaggregated RAN architecture, allowing flexibility and multi-vendor integration. For example, the Radio Intelligent Controller (RIC) may involve third-party applications (xApps) to dynamically control and monitor network behavior, facilitating significant opportunities for multi-party involvement, but allowing potentially untrusted integration with the RAN. In this paper, we propose, design and evaluate *ORANClaw*– a structure aware, man-in-the-middle fuzzing framework that takes full control over the E2 interface between the xApps and the RIC, and systematically mutates or duplicates packets communicated via this interface to disrupt the behavior of the base station (gNB). *ORANClaw* takes into account the structural and semantic constraints while systematically mutating the packets. Furthermore, it optimizes the mutation strategy based on the coverage of explored state transitions. We have implemented *ORANClaw* and evaluated it with FlexRIC, O-RAN SC RIC, OpenAirInterface and ns-3 simulator. In total, *ORANClaw* has discovered 66 unique bugs (seven CVEs already assigned): 28 in FlexRIC, one in O-RAN SC RIC and 37 in the gNB implementations of OpenAirInterface and ns-3. Our evaluation also reveals that structure and semantic-aware mutations within *ORANClaw* are key factors in revealing these bugs. Overall, *ORANClaw* provides an open platform to automatically validate both the RIC and gNB implementations via xApp manipulations.

1 Introduction

Traditional Radio Access Networks (RANs) were typically built as monolithic, vendor-specific systems, limiting flexibility, scalability, and innovation. To address these limitations, the 3rd Generation Partnership Project (3GPP) introduced a disaggregated architecture for 5G RAN in Release 15, separating the base station into three distinct components: the Central Unit (CU), Distributed Unit (DU), and Radio Unit (RU). The Open RAN (O-RAN) standard fundamentally shifted this paradigm by disaggregating RAN functions across standardized interfaces, enabling multi-vendor interoperability and introducing programmable control through the Radio Intelligent Controller (RIC). In particular, the RIC architecture allows third-party applications (xApps) to dynamically optimize network behavior via real-time control and monitoring capabilities through the E2 Application Protocol (E2AP). While this shift from closed, monolithic systems to open, disaggregated architectures enhances flexibility and innovation, it also expands the attack surface [13, 42] and introduces new security challenges that remain insufficiently explored in current research.

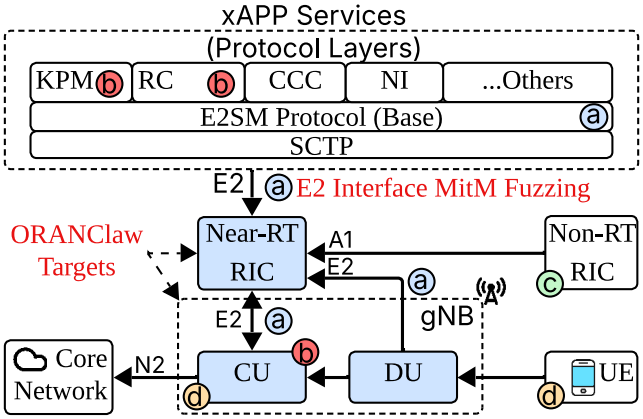

Figure 1: Position of ORANClaw w.r.t related work: a) *ORANClaw* (E2 interface fuzzing) targeting Near-RT RIC and gNB. b) Malicious gNB targeting Near-RT RIC [42], c) Malicious Non-RT RIC targeting Near-RT RIC [40], d) Malicious gNB targeting Core Network [5].

In this paper, we propose *ORANClaw* to systematically investigate a novel attack surface that targets the RAN via third-party and potentially untrusted xApps (see Figure 1ⓐ). *ORANClaw* is a structure-aware and man-in-the-middle (MITM) fuzzing framework, which is placed between the xApps and the RIC (i.e., the E2 interface), as illustrated in Figure 1ⓐ. This allows *ORANClaw* to have full control over the packets through E2 interface and all of its E2SM services, hence acting like a potentially malicious/vulnerable xApp. An appealing feature in the design of *ORANClaw* is that it does not require any access to the RAN source code, thus, facilitating opportunities for testing commercial ORAN stacks. Earlier work on structure-aware fuzzing was targeted to Core Network (see Figure 1ⓓ) and is not directly applicable to our ORAN threat model. In addition, its implementation is unavailable [5] and therefore, extending such implementation remains infeasible in practice. In the O-RAN space, prior works have targeted testing the near real-time RIC (Near-RT RIC) via malicious gNB (see Figure 1ⓑ), or malicious Non real-time RIC (see Figure 1ⓒ). None of these prior works target the gNB or generalize to fuzz all E2SM services (see Figure 1ⓑ). Moreover, unlike *ORANClaw*, prior work on testing Near-RT RIC [42] requires access to the source code. Finally, the works illustrated in Figures 1ⓒ-ⓓ are only based on specific, handcrafted test cases and do not represent a fuzzing framework. *To*

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the best of our knowledge, ORANClaw is the first fuzzing methodology that targets the gNB via systematic manipulation of E2 interface packets that generalizes to all E2SM services.

ORANClaw leverages a benign packet trace to capture the reference state machine of E2AP protocol and xApp Service Models. Then it employs type-aware and constraints-aware mutation strategies for different ASN.1 constructs (integers, bitstrings, sequences) to preserve distinct message structures and semantic constraints. For instance, the Key Performance Measurement (KPM) service model defines specific measurement report formats that must adhere to precise encoding rules. Once the mutated packet is forwarded to the RIC (see Figure 1(e)), *ORANClaw* monitors the communication behaviour in the RAN and reconstructs the state machine on-the-fly to identify new behaviour. This is then used as a feedback to guide the mutation strategies in subsequent fuzzing session. In a nutshell, *ORANClaw* not only implements systematic mutations to carefully modify packets with certain structural and semantic features, but it also enables capabilities to find deeply rooted bugs in specific components of the RAN through stateful coverage of the E2AP protocol and xApp Service Models behaviour.

ORANClaw is not tied to a particular implementation of RAN since it focuses on the E2AP interface between xApp and RIC. Moreover, *ORANClaw* can also be easily extended by changing the xApp and collecting its normal communication behaviour. After providing some background information (Section 2), we make the following contributions:

(1) We present *ORANClaw*, a structure-aware and MITM fuzzing framework that systematically explores the stateful behavior of xApp E2SM service models and E2AP protocol (Section 3).
(2) We implement *ORANClaw* and evaluate it with FlexRIC [35], O-RAN SC RIC [21], OpenAirInterface (OAI) [20] and ns-3 [18]. Our implementation is open source and available for community research (Section 4).
(3) We discover a total of 66 unique bugs (crashes): 19 bugs in the O-DU and O-CU-UP components of OAI in 5G SA configuration, 28 bugs in FlexRIC and its internal libraries (i.e., iApp/SDK), one bug in O-RAN SC RIC I-Release and 18 bugs in the ns-3 5G NSA network simulator. All bugs are reported to the developers and **7** CVEs are already assigned.
(4) We compare *ORANClaw* with alternative strategies and we show that *ORANClaw* finds 13 more unique bugs in comparable fuzzing sessions.

After discussing the limitations of our framework and the closely related works (Section 6 and Section 7), we conclude in Section 8.

2 Background

O-RAN Alliance proposed an open radio access network (RAN) architecture (illustrated in Figure 2) that enables interoperability across multi-vendor environments (i.e., Split 7.2x [23]). This design follows a service-based architecture, in which network functions are implemented as independent microservices. These services, in turn, communicate over standardized interfaces, allowing them to be developed, deployed, and scaled independently on general purpose software and hardware. Beyond this disaggregation, O-RAN introduces two intelligent control entities i.e., the *Near-Real-Time RIC* (Near-RT RIC) and the *Non-Real-Time RIC* (Non-RT RIC) which enable automated, data-driven optimization and management of RAN behavior in response to changing network conditions.

Radio Intelligent Controllers (RIC): The *Near-Real-Time RIC (Near-RT RIC)* has latency requirements typically between 10 ms-1 s [19] and manages dynamic RAN functions e.g., load balancing, interference mitigation, and handover decisions. It interacts with the Central Unit (CU) and Distributed Unit (DU) for precise control. The Near-RT RIC hosts modular *xApps* that implement control and optimization algorithms; these xApps can be developed, deployed, and updated independently, allowing operators to adapt RAN behavior to real-time conditions.

The *Non-Real-Time RIC (Non-RT RIC)*, hosted in the O-RAN Service Management and Orchestration (SMO) framework, operates at a higher abstraction level, handling policy configuration, service orchestration, and long-term optimization. It orchestrates, manages, and automates procedures to monitor and control RAN components across the network and runs *rApps* that analyze aggregated data, produce strategic insights, and set policies that guide the Near-RT RIC and other RAN elements. Through AI-driven analytics the Non-RT RIC enables closed-loop automation for continuous optimization. This is accomplished via the A1, O1 and O2 interfaces.

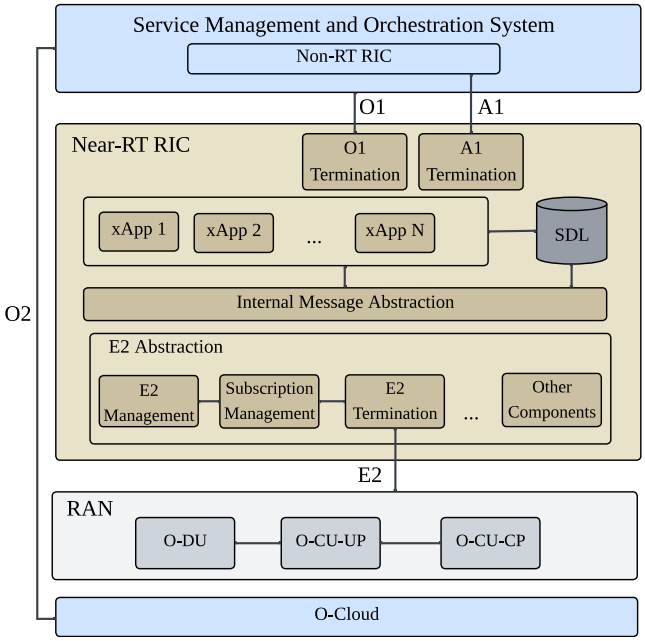


Figure 2: 5G O-RAN Architecture

E2 Interface: The E2 interface (between Near-RT RIC and RAN) operates through the *E2 Application Protocol* (E2AP), which exposes essential RAN internals to enable intelligent control and monitoring [22]. Through the E2 interface, external applications (xApps) running on the Near-RT RIC can continuously monitor network conditions and execute real-time operations. This is facilitated by *E2 Service Models* (E2SM), which define standardized interaction patterns for specific operations such as reporting performance metrics and issuing control commands (e.g., *KPM, RC, etc.*).

Open Source Implementations: The O-RAN Software Community (O-RAN-SC) [21], in collaboration with the Linux Foundation, provides reference implementations for both Near-RT and Non-RT RIC platforms, containerized with Docker and orchestrated via Kubernetes. In particular, the OSC RIC Platform offers full xApp lifecycle management and E2 interface support; FlexRIC [35] delivers a lightweight, easily deployable alternative enabling multi-vendor E2 integration; and ColO-RAN [32] extends the OSC Near-RT RIC, while μONOS RIC from the SD-RAN project [34] emphasizes software-defined RAN capabilities. Non-RT RIC implementations typically enable policy-based control and AI/ML model inference, supported by orchestration frameworks such as ONAP [16], which integrate policy management and RIC control workflows.

Motivation & Threat Model: The O-RAN architecture (Figure 2) introduces new attack surfaces that require careful security analysis. Existing research has primarily examined threats from malicious RAN nodes (e.g., gNB) targeting the RIC [42] or malicious UEs compromising the core network [5]. However, the open nature of O-RAN's xApp ecosystem presents a different and underexplored threat vector: malicious xApps may directly target RAN components (e.g., O-DU, O-CU) and disrupt the network (Figure 1). This risk is recognized by O-RAN Alliance security documents. WG11 identifies threats such as T-NEAR-RT-01/02 (unauthorized access and malicious behavior in the Near-RT RIC) and T-xAPP-01 (malicious xApps) [24, 25]. These reports stress two principles: the Near-RT RIC must not trust data received from the RAN, and strong access control and xApp isolation are required to prevent one xApp from affecting others. Threats T-E2-01 and T-E2-03 further highlight that malformed or injected E2 messages can degrade or disrupt RAN services. To assess this threat in practice, we design an automated testing, in which adversarial xApps systematically inject malformed E2 messages to evaluate RAN robustness. Intuitively, xApps resemble mobile app–store applications and are often produced by third-party vendors. While easy to deploy, such applications may be untrusted or insufficiently tested. *ORANClaw* investigates whether such untrusted xApps can disrupt gNB availability via the E2 interface when deployed within the Near-RT RIC.

3 Design of *ORANClaw*

We discuss key components of *ORANClaw* (Figure 3) below:

3.1 MiTM Proxy

We employ a man-in-the-middle (MiTM) approach to intercept and forward packets exchanged between an xApp and the near-RT RIC. *ORANClaw* implements the component `SCTP Proxy` (as shown in Figure 3) using an SCTP socket in a client-server architecture. Specifically, the `Server` component listens for incoming connections from the xApp, typically on the default E2AP port (i.e., 36422), while the `Client` component initiates a new connection to the actual near-RT RIC. Since different RIC implementations may use different ports (e.g., srsRAN + FlexRIC uses port 34621), *ORANClaw* allows configuring the outbound connection to match the RIC's expected port. This setup enables *ORANClaw* to transparently relay SCTP messages between the xApp and RIC without requiring changes to either party ***(step*** 1 in Figure 3). Furthermore, it enables recovery from any connection failures by either the xApp or the RIC. Using an SCTP socket closely mirrors the real-world communication setup between the near-RT RIC and E2 nodes in production environments, where E2AP messages are transported over SCTP as mandated by the O-RAN specifications [22]. This makes our MiTM compatible with implementations like FlexRIC, which rely on SCTP for E2 message exchange and proprietary E42 messages that adhere to E2AP. From the perspective of an xApp, this communication is entirely transparent. Since xApps typically interact with the RIC through abstracted messaging interfaces and are unaware of the underlying transport details (e.g., SCTP over E2AP), they operate under the assumption that they are directly exchanging information with the RIC. In reality, even if a Man-in-the-Middle (MiTM) entity is inserted between the RIC and the E2 nodes, the xApp continues to function normally sending messages and receiving responses as if the original communication path remained intact. This transparency is crucial for our MiTM-based analysis and fuzzing setup, as it allows us to manipulate, and re-inject E2AP messages without disrupting the logical flow of information perceived by the xApp. Furthermore, if mutations are not desired, the proxy can seamlessly switch to a passive mode, simply forwarding packets between the RIC and the E2 node to preserve baseline communication behavior with a minimal overhead (detailed in Section 5.2).

3.2 Specification Agnostic

ORANClaw is designed to be specification-agnostic, supporting a wide range of E2 Application Protocol (E2AP) versions and E2 Service Models (E2SMs) defined by the O-RAN Alliance [22]. This flexibility is crucial, as different RAN and RIC implementations adopt varying versions of these specifications depending on vendor requirements or research goals. For instance, a setup involving FlexRIC and OpenAirInterface (OAI) gNB may utilize E2AP v2.03 and E2SM-KPM v2.03, whereas a simulation environment like the ns-O-RAN module in ns-3 may adopt older or divergent specifications such as E2AP v1.01, E2SM-KPM v3.00, or E2SM-RC v1.03.

ORANClaw does not rely on hardcoded and/or static message formats; instead, it supports a dynamic code generation workflow in which the desired versions of E2AP and E2SM ASN.1 specification files are discharged to a code generation tool (e.g., `asn1c` [26]). In particular, the `Encoder` and `Decoder` blocks for messages are generated via such a tool. Thus, even though the ASN.1 specification may evolve in future generations, our design can generate the required encoder and decoder for syntactically correct specification files. As the O-RAN ecosystem continues to evolve with newer versions of its protocol stack, *ORANClaw*'s specification-agnostic design ensures it remains both compatible and adaptable over time. Concretely, whenever a new version of the E2AP or an E2SM (e.g., KPM, RC, etc.) is released, *ORANClaw* can generate the required encoders and decoders using the updated ASN.1 files. However, *ORANClaw*'s internal components including its MiTM proxy and `Fuzzing Engine` (discussed next) remain unaffected. This is because they operate over abstracted message representations (e.g., JSON-formatted JER) that are automatically derived from the active specification. Moreover, the process of generating constraints for valid mutations of messages also remain unchanged, as these constraints are retrieved from the same specification files (see Section 3.3.1).

In summary, *ORANClaw* targets comprehensive fuzzing of both the binary-encoded Service Model payload and the higher-level ProtocolIEs. By targeting both structural and semantic (control-level) layers of the message, *ORANClaw*'s `Fuzzing Engine` broadens the protocol's attack surface and increases the likelihood of uncovering implementation-level vulnerabilities in Service Models as well as in high-level semantics of the E2AP protocol.

Service Model (E2SM) generalizability. Unlike prior tools [42] that rely on manually referencing service models data structures (e.g., KPM, RC), *ORANClaw* operates directly on the E2SM base schema (see label *(a)* in Figure 1), without hardcoding references to protocol fields or headers. This zero-coding, schema-driven design allows *ORANClaw* to support any E2 protocol while preserving the semantics of each E2SM service model. Thus, the design and implementation of *ORANClaw* fuzzing engine is agnostic to vendor-specific extensions or future versions of the E2AP, thus effectively generalizing *ORANClaw* MiTM E2 Proxy.

3.3 *ORANClaw* Fuzzing Engine

E2AP messages are encoded using ASN.1 Packed Encoding Rules (PER), which represent data in a compact, bit-level format optimized for transmission efficiency. Directly manipulating such PER-encoded messages is challenging due to the strict bitwise constraints and complex encoding rules. To facilitate precise and flexible fuzzing without violating the protocol specifications, `Fuzzing Engine` internally translates the PER-encoded messages into a structured JSON representation using JSON Encoding Rules (JER). Upon receiving a raw E2AP message, `Fuzzing Engine` first leverages external ASN.1 parsing tools (i.e., *asn1c*) to decode the PER-encoded data into its JSON (JER) equivalent. This transformation maintains the integrity of all bitwise specifications defined by the E2AP standard, ensuring that any modifications applied at the JSON level remain consistent with the protocol's strict constraints.

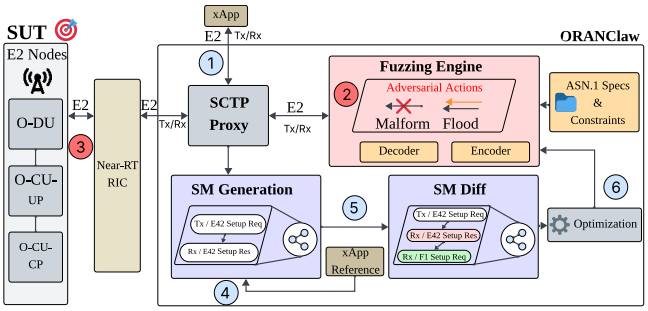


Figure 3: Software Components of *ORANClaw*

After mutations are applied on this JSON representation, the modified message is re-encoded back into PER format before transmission ***(step ❷)***. This decode–mutate–re-encode process allows *ORANClaw* to conduct targeted fuzzing of E2AP fields without breaking protocol compliance or corrupting bit-level details. Furthermore, *ORANClaw* additionally can fuzz and inject invalid mutated values aiming to find issues within the Encoder/Decoder of the RIC's implementation. This is discussed in detail in Section 4.

3.3.1 Type-Aware Fuzzing. To maximize the impact and precision of mutations, *ORANClaw* incorporates a type-aware fuzzing mechanism that leverages the type information embedded in ASN.1 specifications. Rather than relying on blind or random mutations, type-aware fuzzing systematically understands the expected structure, data types, open types and constraints of each field in an E2AP message. This allows the fuzzer to generate inputs that are not only syntactically valid but also semantically plausible, thereby increasing the probability of traversing deeper into the protocol logic and triggering non-trivial behaviors. With this approach, *ORANClaw* respects field-specific properties such as permissible ranges for integers, valid enumeration sets, and string length bounds (as shown in Table 2), ensuring that mutated messages remain within the expected domain of each type. This minimizes early rejections and enables effective exploration of the protocol state space.

Additionally, *ORANClaw*'s `Fuzzing Engine` respects and enforces field-specific constraints derived from ASN.1 specifications during mutation. These constraints include value ranges for integers, size limits for sequences or strings, and required presence conditions for optional fields. By holding such semantic boundaries, constraint-aware fuzzing ensures that generated mutations do not violate basic protocol invariants, allowing the mutated messages to be accepted and parsed by downstream components (see Table 2). This process is agnostic to the version of the Service Model (e.g., KPM v1, KPM v3), and *ORANClaw* automatically extracts these constraints during preprocessing, which starts from the E2SM base layer. After mutation, the payload is re-encoded into its PER (Packed Encoding Rules) format and converted into a hexadecimal string, which replaces the original message content and is subsequently forwarded to the RIC by the `Client` component ***(step ❸*** in Figure 3).

`Fuzzing Engine` also applies mutations directly to the top-level Protocol Information Elements (*ProtocolIEs*), which encode the high-level control semantics of the E2AP protocol. These are structurally distinct from the embedded Service Model (E2SM) payloads carried within specific IEs, although both are defined in ASN.1 and encoded using the same encoding rules. These mutations typically target simpler, non-nested fields that define control parameters, message types, and identifiers. While this "top-level" JSON does not include the Service Model payload, it encapsulates essential structural elements such as the *procedureCode* (e.g., *RICSubscriptionRequest*) and its associated *ProtocolIEs*, which are critical for correct message interpretation and protocol flow control. *ORANClaw* employs mutations to these fields, while respecting schema-derived constraints to ensure syntactic validity.

3.3.2 Mutation Strategy and Field Selection. We heuristically guide the selection and mutation of message fields as follows:

- **Constraint-Driven Mutation:** When detailed constraints are available from the loaded JSON schema files, they guide the generation of random yet valid values for fields such as enumerated types, bounded integers, and patterned strings. This constraint-aware approach helps keep mutations within semantically meaningful boundaries, thereby increasing the chances of uncovering subtle protocol-handling bugs while avoiding immediate rejections due to format violations.
- **Field Importance Weighting:** Initially, the fuzzing process assigns random weights for mutating different message types

and fields. However, these weights are iteratively refined based on the impact of mutations (e.g., finding additional state transitions). We discuss details in Algorithm 1.

- **Payload Complexity Awareness:** The fuzzing engine also targets nested E2SM payloads, which often encapsulate semantic structures such as measurement definitions, triggers, and subscription filters. For example, *ricActionDefinition* is a specific command or instruction that the RIC sends to a Radio Access Network (RAN) node when subscribing to a specific RAN function, and/or to modify its behavior or retrieve information (i.e., E2SM-KPM, that defines the measurement granularity or measurement object IDs). These payloads tend to be more complex and varied.

3.3.3 State Machine Generation and Diff. To capture the runtime behavior of the RAN control interface under fuzzing conditions, we employ an automated state machine generation pipeline. Each session between the xApp and the RIC is transparently intercepted by our MITM proxy, which records the entire interaction into a `.pcapng` file via `TShark` [41]. Once a session concludes, the proxy automatically invokes the `SM Generation` tool on the captured traffic to generate the protocol state machine ***(step*** 4 in Figure 3). In particular, to generate such a state machine, we provide a few rules to identify packet types and protocols along the interfaces within the gNB (i.e., E1 and F1); and between the gNB and the RIC (i.e., E2) from the traffic. Using these rules, we then derive the relevant message sequences and their abstract states, reflecting a behavioral model for a given session. While these rules involve manual input, it is fairly flexible and easy to derive for someone working on the protocol, as shown in prior works [10, 11]. Indeed, within *ORANClaw*, we only needed six rules for generating the state machine (see Appendix B for details).

Based on the intuition mentioned in the previous paragraph, we track protocol deviations or behavioral anomalies introduced by fuzzing. To this end, we generate a differential model by comparing the current session's state machine against a pre-computed baseline model derived from a benign, non-fuzzed execution of the same xApp (i.e., `xApp Reference` in Figure 3). This is performed by invoking `SM DIFF`, which generates a differential model (in the form of a JSON-formatted diff file) that highlights *Added* and *Removed* states (represented by green and red boxes, respectively, in Figure 3), as well as the transitions between them ***(step*** 5**)**. Such a differential model allows us to determine whether a mutated input had an effect, such as triggering new message exchanges and/or activating previously unobserved protocol behaviour. Finally, irrespective of whether the `Optimization` block is enabled or disabled during fuzzing sessions, *ORANClaw* workflow continues iteratively ***(step*** 6**)**. In the subsequent section, we discuss the utility of this differential model within the fuzzing workflow.

3.3.4 Overall Fuzzer workflow. Algorithm 1 presents the overall workflow for *ORANClaw* fuzzing that orchestrates the interaction between packet interception, mutation, and optimization. The algorithm initializes the fuzzing environment by loading ASN.1 constraints C, setting up the fuzzing engine $\mathcal{F}$, and creating the genetic optimizer population $\mathcal{G}$ with initial population (lines 2-7).

During operation, the `SCTP Proxy` intercepts packets $\mathcal{P}$ from xApps and applies probabilistic fuzzing (exemplified by random() $< P_{\text{fuzz}}$ in Algorithm 1) for a supported packet (i.e., packets at E2 interface). When fuzzing is triggered, the current weight vector $\mathbf{w}i$ guides the location of mutation. In particular, $\mathbf{w}i$ is a vector of weights $[w_{\text{SM}}, w_{\text{json}}, w_{\text{both}}]$ where w_{SM} and w_{json} capture the probabilities of mutating either the service model or the top-level JSON representation (e.g., *RIC-SubscriptionRequest*), respectively, whereas w_{both} denotes the probability to mutate both the service model and top-level JSON. Intuitively, $\mathbf{w}i$ keeps track of the *importance* of the different parts of messages (i.e., the service model, the top-level JSON representation or both) from the viewpoint of fuzzing. In other words, if mutating service models predominantly leads to functional anomalies (e.g., crashes) as compared to mutating other parts of the message, then the genetic algorithm embodied within *ORANClaw* naturally assigns more weights to w_{SM}. While this idea could be extended to any number of message fields trivially, the implementation of *ORANClaw* sticks to the current $\mathbf{w}i$ for the sake of simplicity. Once the location of mutation (i.e., service model, top-level JSON or both) is chosen in line with $\mathbf{w}i$, the fuzzing engine selects $\mathcal{F}_{\text{fields}}$ for up to $M \leq M_{\max}$ mutations. These fields are then leveraged to apply type-aware mutations $\mathcal{F}.\texttt{mutate}(\mathcal{P}, \mathcal{F}_{\text{fields}}, C)$ to produce the mutated packet $\mathcal{P}'$.

The algorithm supports optional packet duplication when enabled and random() $< P_{\text{dup}}$, sending $D \leq D_{\max}$ copies of $\mathcal{P}'$ to stress-test the target RIC's handling of repeated messages. To enable adaptive mutations, we record mutation traces and update fuzzing statistics (e.g., found crashes or additional state transitions) for fitness evaluation when $\mathcal{P}' \neq \mathcal{P}$. In particular, session boundaries are detected through timeouts or natural termination points, at which the system extracts protocol states $\mathcal{S}$ from logs and computes the cost function $c(\mathbf{w}_i)$ based on observed protocol transitions and applied mutations (see lines 34-36 in Algorithm 1). Finally, the function $\mathcal{G}.\texttt{update}(c(\mathbf{w}_i))$ evolves the population and a new fuzzing session starts (line 13 in Algorithm 1). This continuous feedback loop enables adaptive fuzzing strategy refinement across sessions while maintaining comprehensive logging through session recording and differential model generation for subsequent analysis.

3.3.5 Optimization Objective (Cost Function). The state machine embodied in *ORANClaw* (see "SM Generation" in Figure 3) captures states (message types and direction) that directly correspond to attacker-controlled payloads over E2 interface. Intuitively, *ORANClaw* aims to diversify mutations on such attacker controlled payloads. This is to discover unexpected or erroneous behaviour. As a result, the optimizer aims to maximize the transitions in the state machine. Concretely, we design the cost function as follows:

$$c = \beta \cdot \frac{T}{\tau} + \mu \cdot M \tag{1}$$

where:

- T: Number of observed transitions during a fuzzing session.
- τ: Normalization constant representing the number of transitions in the reference state machine model (i.e., the behavioral model in the absence of fuzzing).
- M: Total number of mutations applied to achieve the observed state changes and transitions.
- β, μ: Hyperparameters weighting the relative importance of transition exploration and penalty for mutation of fields, respectively.

Algorithm 1 Main steps of *ORANClaw*

```
procedure ORANClaw_Fuzzing
    C ← Load Constraints
    F ← Initialize Fuzzing Engine
    G ← Initialize GeneticFuzzerOptimizer
    Start MiTM proxy and capture interface
    ▷ Initialize population with uniform Dirichlet
    W ← {w_i ~ Dirichlet(1) : i = 1, ..., N}
    while proxy is running do
        ▷ Run Fuzzing session for each individual
        for each w_i ∈ W do
            c(w_i) ← ORANClaw_Session(w_i)
        ▷ Update population
        W ← G.update([w_i, c(w_i)]_{1...N})

procedure ORANClaw_Session(w_i)
    while true do
        Receive: P ← packet from xApp
        if random() < P_fuzz and P is supported then
            w_i ← current weights from G
            M ← #mutations sampled s.t. M ≤ M_max
            F_fields ← select fields to mutate using w_i
            P' ← F.mutate(P, F_fields, C)
            if P' ≠ P then
                Record mutation trace for fitness
                Update fuzzing stats
            if duplication enabled and random() < P_dup then
                D ← random count ≤ D_max
                Send D copies of P' to target                ▷ Duplicate
            else
                Send P' to target                ▷ Send fuzzed packet once
        else
            Send original P to target                ▷ No fuzzing applied
        if session ends or timeout then
            Extract protocol states S from logs
            Save session, generate state machine diff
            Compute cost c(w_i)                ▷ Cost Function
            return c(w_i)
```

Intuitively, the term $\beta \cdot \frac{T}{\tau}$ elevates the value of the cost function when crashes or functional anomalies occur, as such events typically increase the number of new messages exchanged between the RIC and the gNB, leading to several new transitions in the state machines. μ is typically a small value (set to 0.1 in *ORANClaw*), which ensures that the increased number of mutations does not have a significant impact on the value of the cost function. This ensures that the optimizer does not disproportionately favor populations with higher mutations.

4 Implementation and Experimental Setup

The *ORANClaw* implementation is shown in Figure 3. Our implementation is built around a custom MitM `SCTP Proxy` that intercepts SCTP traffic between xApps and the RIC, which then forwards communication to E2 Nodes. This architecture enables manipulation and analysis of E2AP protocol messages during runtime.

ORANClaw is developed using Python 3.12 (1261 LoC) and runs on a machine with an AMD Ryzen 7 5800H CPU @ 4.46GHz, with 64GB of RAM for a single session. Additionally, concurrent fuzzing sessions used in the experiments for **RQ1** (discussed in Section 5.1), were run on a machine with an AMD Ryzen Threadripper PRO 7985WX CPU @5.37GHz and 512GB of RAM.

MitM `SCTP Proxy`: It implements a client-server architecture using Python and integrates the following components:

- *Core Communication Layer:* `pysctp` [30] is used for low-level SCTP socket communication, establishing bi-directional connections between xApps and the RIC platform.
- *Traffic Analysis:* We leverage `TShark` to capture and log intercepted packets, generating *pcapng* files for post-fuzzing session analysis.
- *Message Processing: ORANClaw* processes E2AP messages using two complementary approaches: the Objective Systems ASN1C [26] compiler generates C++ runtime libraries from ASN.1 specification files, while *asn1tools* provides further Python-based encoding/decoding capabilities on the JER representation.

Bidirectional Message Conversion: As discussed in Section 3.3, *ORANClaw* implements a bi-directional conversion enabling PER encoding/decoding allowing proper communication between the xApp and the RIC:

(1) `Decoder`: This component (127 LOC) receives hexadecimal streams representing PER-encoded E2AP messages and converts them to JER (JSON Encoding Rules) format. The decoder extracts protocol data from the binary PER representation and generates human-readable JSON structures (JER encoding) for mutation.
(2) `Encoder`: This component (156 LOC) processes the JER-compliant E2AP message structures and converts them back to PER format (possibly after mutation). This enables modified messages to be transmitted properly.

Both the `Encoder` and `Decoder` components are custom-built wrappers around C++ functions, which are automatically generated by the *asn1c* compiler. Moreover, such a *decode-modify-encode* workflow introduces minimal overhead: it is within the typical RIC–xApp communication interval ($1ms$ to $1s$) [19].

A challenge arises when processing certain E2AP message types (i.e., *RICSubscriptionRequest*). These messages contain Service Model definitions (e.g., E2SM KPM and E2SM RC) within fields such as *ricActionDefinition*. The standard *asn1c* compiler cannot decode these complex PER payloads that define Service Model semantics. To overcome this limitation, we leverage the *asn1tools* Python library specifically for Service Model encoding and decoding. This ensures consistency by using the same ASN.1 specification files.

Radio Intelligent Controller: Due to the complexity and overhead of deploying the full O-RAN Software Community (SC) Near-RT RIC, which involves orchestrating multiple containers (e.g., E2 Termination, Submgr, Rtmgr, and more), we adopt FlexRIC for most of our experiments. FlexRIC is a modular alternative that implements the essential E2 interface components with minimal dependencies, making it significantly easier to deploy, and integrate for testing. This simplicity is particularly advantageous in our context, where deep packet inspection, fuzzing instrumentation, and packet mutations are required. By using FlexRIC, we adhere to the E2AP protocol while avoiding the deployment complexity

of O-RAN-SC RIC. To validate the generalizability of *ORANClaw* beyond FlexRIC, we also evaluate it with the full O-RAN SC RIC deployment, as discussed in Section 5.5.

System under test (SUT): *ORANClaw* primarily evaluated two open-source RAN implementations:

The first deployment leverages the OpenAirInterface (OAI) gNB stack [28], integrated with FlexRIC as the near-Real-Time RIC platform to support xApp deployment. The setup is completed with Open5GS [27] as the core network (5GC), forming a full end-to-end 5G system. The OAI stack used in this setup use the disaggregated gNB architecture aligned with 3GPP and O-RAN specifications. Specifically, it includes the O-DU, O-CU-UP, and O-CU-CP functional components, instantiated through the `nr-softmodem` application. These components communicate via standardized interfaces F1 (between O-DU and O-CU) and E1 (between O-CU-CP and O-CU-UP) supporting a realistic split architecture. For testing, two simulated UEs are deployed within the environment (i.e., *nr-uesoftmodem*), interacting with the gNB and core network. The entire setup is deployed within a containerized environment to facilitate the testing and reproducibility.

We also deploy a 5G NSA setup via the *ns3-mmwave-oran* module (`scenario-zero.cc`) [37]. The experimental setup has three simulated gNBs (`MmWave Nodes`), one eNB (`Lte Nodes`), and five UEs with random-walk mobility. FlexRIC runs the xApps, and uses the Evolved Packet Core (EPC) as its core network. The gNB, instantiated via *MmWaveEnbNetDevice*, implements a monolithic stack without real CU/DU or control/user-plane separation. Although not a fully disaggregated deployment, we argue that this abstraction is sufficient for evaluating most of the xApp-driven control logic.

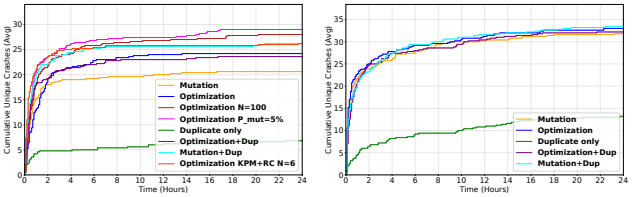
5 Results

We answer the following research questions (RQs) to evaluate the effectiveness and efficiency of our *ORANClaw*:

5.1 RQ1: How effective is *ORANClaw* in finding bugs?

In Table 3, we showcase the effectiveness of *ORANClaw* in finding bugs in open source implementation (Implementation column) of ORAN components such as near-RT RIC (FlexRIC), O-DU, O-CU (OAI) and monolithic gNB and eNB (ns-3). To this end, each target implementation is evaluated against *ORANClaw* five times for a period of 24 hours with the same optimization parameters (i.e., β, μ) to keep consistency during the evaluation. The results are classified by their *Vulnerability* type, *General Cause*, affected ORAN *Component* and *Threat*.

Furthermore, to highlight *ORANClaw* effectiveness over time, we evaluate different configurations of *ORANClaw* (i.e., Optimization, Mutation only, Duplication only, etc.) by running five fuzzing sessions for each configuration and we show the average of the cumulative unique bugs in Figures 4(a)-(b). Additionally, in the OAI setup we performed a sensitivity analysis of the cost function in Eq. 1 by increasing the population size (i.e., $N = 100$), while keeping the rest of the experiments at $N = 6$. We also reduced the mutation hyperparameter P_{mut} to 5%, compared to the 30% used in the other experiments. Finally, we evaluated an additional xApp test case in FlexRIC that combines the KPM and RC service models to demonstrate the generalizability of E2SM.



(a) OpenAirInterface **(b) ns-3**

Figure 4: Average of cumulative bugs with respect to time.

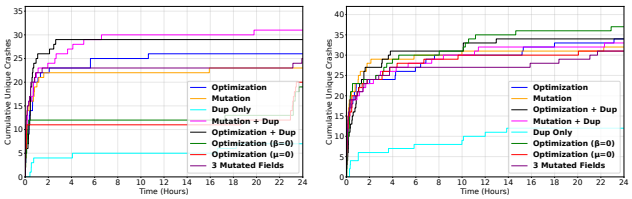
Overall, *ORANClaw* and its variations were able to identify a total of **66** vulnerabilities: **19** issues in the O-DU, O-CU-UP components of OAI in 5G SA configuration, **28** bugs in FlexRIC and its internal libraries (i.e., *iApp/SDK*), 1 bug in O-RAN SC I release and **18** bugs in the ns-3 5G NSA network simulator. Interestingly, *ORANClaw* also uncovered a DoS issue in OAI related to the E1 interface (*VulnOAI-11*), which is not directly connected to xApps nor the near-RT RIC. Such vulnerability is not triggered by a malformed message, but rather from the O-CU-UP's inability to recover properly after a previous fuzzing session that led to a crash. Such results demonstrate *ORANClaw*'s effectiveness of finding bugs in several ORAN components that are indirectly connected to xApps (gNB, CU, DU), highlighting the importance of securing the RAN against malicious xApps. We note that 63 out of 66 cases (Table 3) do not require continuous adversarial presence; a single malformed message triggers the fault. In these cases, the system fails to recover without manual restart (persistent DoS). All issues were reported to the developers of the target implementation, with many bugs already having their corresponding CVEs assigned.

5.2 RQ2: How do components of *ORANClaw* contribute to its overall effectiveness?

To investigate the contribution of our fuzzing strategies, we evaluate variants of *ORANClaw* solely employing mutation, duplication, optimization and a combination of such. Furthermore, for the optimization, parameters of the cost function such as β and μ are disabled to study their impact throughout the fuzzing process. The results of our evaluation for different 5G stacks (OAI and ns-3) are highlighted in Figures 5(a)-(b).

We further study the effect of state-machine completeness on *ORANClaw*'s performance (Figure 6). Incomplete state-machine traces (green, olive, dark red) sometimes match or outperform the fully reconstructed trace (blue) in terms of unique crash discovery (Figure 5(b)). This suggests that partial traces can encourage exploration of alternative or out-of-order transitions that the complete trace constraints, enabling the fuzzer to reach different states. Thus, completeness is not always strictly beneficial in some cases, a less prescriptive trace can increase coverage of unexpected behaviours.

In summary, variants *optimization + duplication* and *mutation + duplication* consistently find most bugs across the evaluated RAN implementations. Moreover, *mutation + duplication* often matches or even outperforms *optimization + duplication*. This highlights that semantic- and constraint-aware mutations are the critical factors at



(a) OpenAirInterface (b) ns-3

Figure 5: Unique Crashes w.r.t Time and fuzzing strategies

exploiting protocol fields to reveal bugs. However, the variant with the number of transitions disabled ($\beta = 0$), performs better with ns-3 since majority of ns-3's vulnerabilities are located in the early states of the communication (xApp registration). This is because the full optimization variant incentivizes exploration of bugs in further protocol transitions, making it less effective to find bugs early in the state machine. Nonetheless, we argue that leveraging protocol transitions as cost function is appropriate since it targets a broader use case for RAN service and can find most of the bugs in the initial hours of the fuzzing session for both evaluated RAN implementations (see Figures 5(a)-(b)).

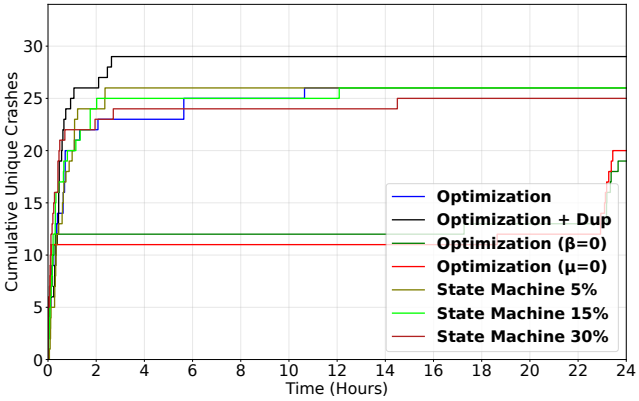


Figure 6: Unique Crashes in OAI w.r.t Time with state machine completeness

5.3 RQ3: How efficient is *ORANClaw*?

In order to fuzz the RAN, *ORANClaw* intercepts E2AP messages via its internal bridge and performs packet processing which includes encoding and decoding of ASN.1 messages. Therefore, to measure the overhead of *ORANClaw* when intercepting through the E2 interface, we measure the time it takes for every packet to enter and leave the E2 bridge over a period of 24 hours. We perform this experiment by enabling and disabling packet processing and showcase the overhead results in the boxpot of Figure 7.

The results show that *ORANClaw* adds around 13*ms* of overhead to the E2 interface when processing packets. This delay is low enough to satisfy the timing requirements (10*ms* to 1*s*) of the E2 interface [19]. This demonstrates the feasibility in employing *ORANClaw* to time-critical RAN interfaces.

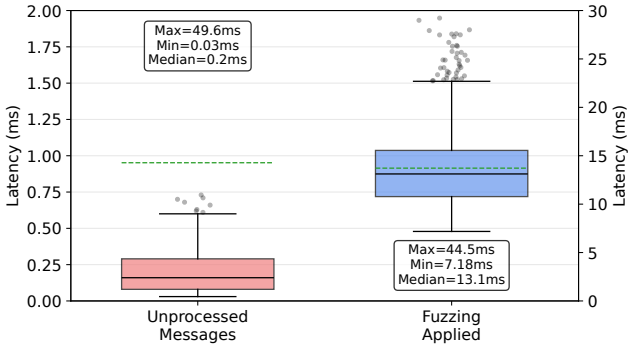


Figure 7: Latency of *ORANClaw*

5.4 RQ4: How *ORANClaw* compares w.r.t state of the art testing tools?

Due to the limited tooling that can mutate ASN.1 fields in the context of ORAN, we evaluate the competitiveness of *ORANClaw* by modifying it to match the fuzzing behaviour of other tools e.g., Boofuzz [29]. This is done by conducting a fuzzing session in the OAI setup with basic random JSON mutations without considering ASN.1 constraints (structure-unaware). Our results show that 87.9% of the messages (904/1029) leads to basic encoding/decoding issues due to mutations, often breaking ASN.1 parsing constraints. Moreover, due to such ASN.1 parsing errors, the session only lasts for six hours as no differences are observed beyond such a timeframe.

Overall, only eight crashes are triggered within the internal message parser of the RIC. Moreover, two other crashes are triggered within the gNB due to failures in the RIC. In contrast, *ORANClaw* finds ∼23 bugs within the same six hours time window (see Figure 4(b)). This shows the effectiveness of *ORANClaw* and its type-aware, constraint-based mutations.

Additionally, we evaluated the 5G fuzzer available in U-Fuzz [36]. However, U-Fuzz is unable to fuzz the E2AP ASN.1 SCTP payloads sent from xApp to the RIC. This is because U-Fuzz relies on Wireshark to decode ASN.1 messages, which, in turn does not support decoding the same version of the ASN.1 E2AP protocol employed in FlexRIC. Therefore, we were unable to evaluate U-Fuzz further due to its limitation in parsing different versions of ASN.1 protocols. We note that *ORANClaw* does not face such an issue and it is robust to different versions of ASN.1. This is because *ORANClaw* leverages compiled code from arbitrary ASN.1 specifications for message encoding and decoding.

5.5 *ORANClaw* Generalizability

To demonstrate that *ORANClaw* does not require fundamental changes to its core fuzzing engine or overall architecture, we evaluated it against the O-RAN SC RIC platform [21] together with the srsRAN gNB [39], demonstrating its deployability across multiple RIC and base station implementations. For this evaluation, we used the O-RAN SC I-Release and the single-binary gNB, following the official tutorial [38]. By simply placing *ORANClaw* in between the E2 Termination and the gNB (see Figure 2), we deployed two virtual machines (VMs): one running the O-RAN SC RIC and another running both the gNB and *ORANClaw*, which we controlled via SSH.

We redirected *ORANClaw*'s `Fuzzing Engine` to target the O-RAN SC RIC instead of the original target (i.e., gNB). After 11 hours of fuzzing, we discovered one vulnerability in the E2 Termination component (**VulnOSCRIC-1**, as shown in Table 3). This demonstrates the generalizability of the MiTM fuzzing engine embodied in *ORANClaw* across multiple ORAN deployment scenarios.

5.6 Example Vulnerabilities

5.6.1 Mutating E2SM-KPM. When deploying *ORANClaw* alongside the OAI implementation, we evaluated the default FlexRIC xApp using the KPM Service Model (i.e., xapp_kmp_moni) that retrieves downlink and uplink metrics from UEs at specified intervals *(granularityPeriod)*. As demonstrated in **VulnOAI-06** (see Table 3), *ORANClaw* successfully decodes the KPM payload's Protocol Information Elements (IEs) and exploits the measurement reporting structure. Specifically, *ORANClaw* mutates the KPM payload by modifying the *measName* field with a malformed structure, while maintaining a valid *measID* format (see Listing 1). This targeted manipulation of E2SM-KPM IEs triggers a parsing error in the O-DU that leads to an assertion failure and crash.

```
        "measInfoList": [
          {
- "measType": ("measName", "DRB.Pdcp...DL"),
+ "measType": ("measID", 5354),
            "labelInfoList": [{"measLabel": {"noLabel":
                "true"}}]
          },
          {
- "measType": ("measName", "DRB.Pdcp...UL"),
+ "measType": ("measID", 5354),
            "labelInfoList": [{"measLabel": {"noLabel":
                "true"}}] }, ...
```

Listing 1: Structure Mutation by ORANClaw in E2SM-KPM

5.6.2 Single-Field Mutation & LTE eNB. *ORANClaw* also mutates single fields (e.g., single byte) within ASN.1 structures. For instance, when mutating the *Criticality* field from *id-RICsubscriptionDetails* ProtocolIE (i.e., "id:30"), a vulnerability is triggered in the eNB from ns-3 Simulator. It processes the mutated RICSubscriptionRequest but its *e2sim* module fails during ASN.1 APER encoding, triggering an assertion error in the channel validation logic (`m_channel`), ultimately crashing the entire eNB with a SIGIOT signal. This implementation vulnerability is noted as **VulnNS-06** in Table 3. In general, this security flaw shows that malformed messages via the E2 interface may propagate through and even disrupt the eNB in a 5G NSA configuration.

```
        {
          "id": 30,
- "criticality": "reject",
+ "criticality": "notify",
          "value": {
            "ricEventTriggerDefinition": "0063",
            "ricAction-ToBeSetup-List": [
```

Listing 2: Criticality Field Mutation in RIC Subscription Request

6 Discussion & Limitations

O-RAN Testing: Our field selection strategy is primarily based on its location (for example, at the service-model level or among top-level JSON fields) rather than more nuanced and fine-grained selection strategy for each individual field. Nonetheless, the open platform and architecture of *ORANClaw* allow design and implementation of more sophisticated selection and optimization strategies for message mutations.

Code/branch coverage: *ORANClaw* is designed as a black-box testing tool and it does not take into account code coverage. While this approach enables testing of proprietary implementations, it may result in lower code coverage. However, *ORANClaw* can be easily extended via standard code instrumentation tools used in greybox fuzzer family [33, 43].

Testing Completeness: Our current implementation focuses on E2AP message fuzzing and does not focus on other O-RAN interfaces such as F1, or Open Fronthaul protocols. In particular, since *ORANClaw* only focuses on mutating the messages captured over the E2 interface, it does not find vulnerabilities that are not caused by these messages. Nonetheless, the systematic mutation strategies implemented over ASN.1 messages and the optimization process based on the differential model are generic. We aim to adapt *ORANClaw* for fuzzing other ORAN interfaces in the future.

Compiler-generated code: *ORANClaw* generates the necessary *C++* components (`Encoder, Decoder`) by discharging ASN.1 specification files to *asn1c* compiler. While such an architecture makes *ORANClaw* robust to future changes in ASN.1 specification, it also creates a dependency on the correctness and completeness of the compiler. In our work, we preferred a compiled version of the Message Encoder and Decoder, as such is more reliable and extensible over manual and hard-coded rules to generate messages.

Manual efforts: *ORANClaw* requires a benign xApp trace file (e.g., *.pcapng*) to establish a baseline for the `SM DIFF module` (see Figure 3) when constructing state machines. Additionally, it also requires a few rules to identify the packet types and protocols. Nonetheless, these rules are easy to be specified with minimal knowledge of the target protocol (E2AP) and wireshark [11].

7 Related Work

Table 1 illustrates the comparison between *ORANClaw* and other related works. In the following, we discuss and position *ORANClaw* with respect to prior works.

Greybox Fuzzing: Automated testing within the RIC environment is challenging due to the nature of the messages defined in ASN.1 structures and the dynamic nature of E2AP protocol. Specifically, existing general fuzzing tools [33, 43] involve source-code instrumentation and they do not focus on stateful protocols. While stateful and generic greybox fuzzing have made some progress in the last few years [4, 31], they are not directly applicable within the RIC environment (see columns *Type* and *Attack Model Applicability* in Table 1). For example, they are based on mutating static seed input (packet sequences), whereas AFLNET specifically requires a response for every request. Mutating static seeds are not appropriate for fuzzing dynamic protocols, as the mutated packets are often invalid and rejected by the target node. In contrast, the mutations

embodied in *ORANClaw* are aware of the ASN.1 structure and it also involves man-in-the-middle fuzzing (at E2) to deal with the complexity of dynamic protocols.

Work	Stateful Execution	Type	Grammar/Structure Awareness	ASN.1 Support	Attack Model Applicability
AFL [43]	❍	Greybox	❍	❍	NA
AFLNET [31]	◗	Greybox	❍	❍	NA
LibFuzzer [33]	❍	Greybox	❍	❍	NA
Frizzer [17]	❍	Blackbox	◗	❍	NA
BooFuzz [29]	❍	Blackbox	◗	❍	Protocol-specific
Peach [7]	❍	Generation-based	◗	❍	Protocol-specific
U-Fuzz [36]	●	Blackbox	●	❍	Protocol-specific
5Ghoul [11]	●	Blackbox	●	❍	UE focused
ORANALYST [42]	❍	Whitebox	●	◗	RIC focused
Berserker [2]	❍	Whitebox	●	●	UE/RAN focused
RANsacked [5]	❍	Whitebox	●	●	Core Network focused
ORANClaw	●	Blackbox	●	●	RAN focused

Table 1: *ORANClaw* vs other testing tools. ●: Full consideration, ❍: Exclusion, ◗: Partial consideration.

Protocol Fuzzing: In contrast to *ORANClaw*, BooFuzz [29] and Peach [7] require manual specification of message grammars (see column *Grammar* in Table 1). Frizzer [17] dynamically instruments the target using the Frida toolkit. However, due to the dynamic instrumentation, it is inefficient and involves reverse engineering to indicate target function address. On the contrary, *ORANClaw* does not involve any reverse engineering or instrumentation. Moreover, it automatically constructs the state machine from the communication captured in E2 interface. Hence, it is suitable to effectively and efficiently fuzz both open- and closed-sourced targets.

Limited ASN.1 support: Efforts have been made to develop tools that aim to generate inputs based on ASN.1 structures and grammar. For instance, prior works targeting the RIC [42] lacks statefulness and it also involves manual and direct implementation of mutation rules on certain ASN.1 structures. As a result, such an implementation is not easily extensible for future evolutions of ASN.1 specification (see column *ASN.1 Support* in Table 1). Furthermore, libprotobuf-mutator [12] enables the generation of ASN.1 messages using protobuf definitions, but it relies on a manual translation from ASN.1 to protobuf. This manual process is particularly challenging when testing O-RAN protocols such as E2AP and E2SM [22], which involve hundreds of complex structures across various types.

RAN-focused fuzzers: Although Berserker [2] can generate valid ASN.1 messages, its RRC focus and stateless execution limit testing scope (see column *Stateful Execution* in Table 1). Other approaches such as RANsacked [5] and prior core/RAN work [8] attack the network only via a malicious UE, while U-Fuzz [36] targets the UE from a rogue base station. Likewise, some efforts target only the near-RT RIC via a malicious base station [42] or the A1 interface [40], narrowing their applicability. By contrast, *ORANClaw* exercises multiple RAN components over the E2 interface without deploying malicious devices, producing practical findings since arbitrary malicious xApps can inject E2 attacks. Prior xApp-triggered testcases [13] lack flexible, automated discovery as O-RAN and ASN.1 evolve; *ORANClaw* addresses this by using existing xApps and an ASN.1-agnostic, automated testing approach.

O-RAN Attacks and Defense: Several works uncover vulnerabilities specific to ORAN and its implementations [9, 15], while other works focus on defense [1, 3, 6, 14]. Instead of competing, *ORANClaw* complements these works by providing a framework that allows packet analysis regardless of the ASN.1 specification. Consequently, future research can build upon *ORANClaw*.

8 Conclusion

In this paper, we comprehensively investigate the attack surface that targets the RAN via third-party and potentially untrusted xApps. To this end, we propose and design *ORANClaw*, which acts as a man-in-the-middle (MITM) fuzzer at the E2 interface between the xApp and the RIC. The MITM fuzzing capabilities embodied in *ORANClaw* allows us to impersonate the behaviors of arbitrary vulnerable and malicious xApps via systematic packet mutations and duplications. An appealing feature of *ORANClaw* is its capability to perform type-aware, constrained mutations that stress test the behavior of the RAN (RIC and the base station) without being rejected early by the packet decoder at the RAN. Besides, *ORANClaw* employs optimizations of mutation strategies via a differential state model to find deeply rooted bugs in the E2AP protocol. The effectiveness of *ORANClaw* is demonstrated by finding 66 unique bugs across widely used RIC (FlexRIC, O-RAN SC RIC) and gNB implementations (OAI and ns3-simulator). While *ORANClaw* is currently applied for fuzzing the RAN, we believe its core fuzzing engine can be extended for fuzzing different segments of the ORAN architecture. This is because *ORANClaw*'s fuzzing engine is based on ASN.1 specification. We hope that the design of *ORANClaw* opens the possibility of security testing for the evolving ORAN architecture at scale. For reproducing our results and advancing research in the area, we have made our code and data available at:

https://anonymous.4open.science/r/ORANClaw-E2-MitM-Fuzzing-AFFC/

9 Research Ethics

All experiments were conducted in isolated environments using O-RAN software with no impact on live networks or user data. Discovered vulnerabilities were reported through coordinated disclosure to strengthen O-RAN security before widespread deployment.

References

[1] Zakaria Abou El Houda, Hajar Moudoud, and Bouziane Brik. 2024. Federated deep reinforcement learning for efficient jamming attack mitigation in O-RAN. *IEEE Transactions on Vehicular Technology* 73, 7 (2024), 9334–9343.

[2] Researchers at KTH. 2021. Berserker: ASN.1-Aware Telecom Protocol Fuzzing. In *Telecom Security Conference*. focus on NGAP, F1AP, XnAP.

[3] Tolga O. Atalay et al. 2023. Securing 5G openran with a scalable authorization framework for xapps. In *IEEE INFOCOM*. IEEE, 1–10.

[4] Jinsheng Ba, Marcel Böhme, Zahra Mirzamomen, and Abhik Roychoudhury. 2022. Stateful greybox fuzzing. In *USENIX Security Symposium*. 3255–3272.

[5] Nathaniel Bennett et al. 2024. Ransacked: A domain-informed approach for fuzzing lte and 5g ran-core interfaces. In *CCS*. 2027–2041.

[6] Shin-Ming Cheng, Bing-Kai Hong, and Cheng-Feng Hung. 2022. Attack Detection and Mitigation in MEC-Enabled 5G Networks for AIoT. *IEEE Internet of Things Magazine* 5, 3 (2022), 76–81.

[7] Peach Tech Community. 2021. Peach Fuzzer Community Edition. https://peachtech.gitlab.io/peach-fuzzer-community/. supports both generation- and mutation-based fuzzing via Peach Pit grammar files.

[8] Dimitri Dessources et al. 2024. In *MILCOM*. 129–134.

[9] Yared Abera Ergu et al. 2024. Efficient adversarial attacks against DRL-based resource allocation in intelligent O-RAN for V2X. *IEEE Transactions on Vehicular Technology* (2024).

[10] Matheus E. Garbelini et al. 2022. BrakTooth: Causing Havoc on Bluetooth Link Manager via Directed Fuzzing. In *USENIX Security Symposium*. USENIX Association, 1025–1042.

[11] Matheus E. Garbelini et al. 2025. 5GHOUL: Unleashing Chaos on 5G Edge Devices via Stateful Multi-layer Fuzzing. *IEEE TDSC* (2025).

[12] Google. 2023. libprotobuf-mutator. https://github.com/google/libprotobuf-mutator. Accessed: 2025-08-02.

[13] Cheng-Feng Hung et al. 2024. Security threats to xApps access control and E2 interface in O-RAN. *IEEE Open Journal of the Communications Society* 5 (2024), 1197–1203.

[14] Cheng-Feng Hung, Chi-Heng Tseng, and Shin-Ming Cheng. 2025. Anomaly Detection for Mitigating xApp and E2 Interface Threats in O-RAN Near-RT RIC. *IEEE Open Journal of the Communications Society* 6 (2025), 1682–1694.

[15] Felix Klement et al. [n. d.]. Endless Subscriptions: Open RAN is Open to RIC E2 Subscription Denial of Service Attacks. ([n. d.]).

[16] Hoejoo Lee et al. 2020. Hosting ai/ml workflows on o-ran ric platform. In *IEEE Globecom Workshops*. IEEE, 1–6.

[17] Dennis Mantz and the Frizzer community. 2025. Frizzer: A Coverage-Guided Black-Box Fuzzer Based on Frida. https://github.com/demantz/frizzer.

[18] Marco Mezzavilla et al. 2015. 5G mmWave module for the ns-3 network simulator. In *Proceedings of the 18th ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems*. 283–290.

[19] Esteban Municio et al. 2023. O-ran: Analysis of latency-critical interfaces and overview of time sensitive networking solutions. *IEEE Communications Standards Magazine* 7, 3 (2023), 82–89.

[20] Navid Nikaein et al. 2014. OpenAirInterface: A flexible platform for 5G research. *ACM SIGCOMM Computer Communication Review* 44, 5 (2014), 33–38.

[21] O-RAN. 2025. O-RAN Software Community. https://o-ran-sc.org/.

[22] O-RAN. 2025. O-RAN Specifications. https://orandownloadsweb.azurewebsites.net/specifications.

[23] O-RAN Alliance. 2023. O-RAN Fronthaul Working Group 4: Control, User and Synchronization Plane Specification. https://www.o-ran.org/specifications. O-RAN.WG4.CUS.0-R003-v11.00, Release 003, Version 11.00.

[24] O-RAN Alliance. 2023. *O-RAN Security Threat Modeling and Remediation Analysis*. Technical Report. O-RAN Alliance WG11. O-RAN.WG11.Threat-Model.O-R003-v06.00.

[25] O-RAN Alliance. 2023. *Security for Near-RT RIC and xApps*. Technical Report. O-RAN Alliance WG11. O-RAN.WG11.Security-Near-RT-RIC-xApps-TR.0-R003-v03.

[26] Objective Systems, Inc. 2024. ASN1C ASN.1 Compiler. https://www.obj-sys.com/products/asn1c/. ASN.1 C/C++ Compiler and Runtime Libraries.

[27] Open5GS Project. 2024. Open5GS: Open Source Implementation of 5G Core and EPC. https://open5gs.org/. Accessed: 2025-08-05.

[28] OpenAirInterface Software Alliance. 2024. OpenAirInterface 5G Software Alliance. https://openairinterface.org/. Accessed: 2025-08-05.

[29] Josh Pereyda and the BooFuzz community. 2023. BooFuzz: Network Protocol Fuzzing for Humans. https://github.com/jtpereyda/boofuzz.

[30] Elvis Pfutzenreuter. 2022. pysctp: Python bindings for the SCTP transport protocol. https://pypi.org/project/pysctp/. Maintained by Benoit Michau; accessed August 5, 2025.

[31] Van-Thuan Pham, Marcel Böhme, and Abhik Roychoudhury. 2020. AFLNET: A Greybox Fuzzer for Network Protocols. (2020), 460–465.

[32] Michele Polese et al. 2022. ColO-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms. *IEEE Transactions on Mobile Computing* 22, 10 (2022), 5787–5800.

[33] LLVM Project. 2021. libFuzzer: In-process, coverage-guided, evolutionary fuzzing engine for LLVM. https://llvm.org/docs/LibFuzzer.html.

[34] Onos Project. 2025. SDRAN-in-a-Box for SD-RAN project. https://github.com/onosproject/sdran-in-a-box.

[35] Robert Schmidt, Mikel Irazabal, and Navid Nikaein. 2021. FlexRIC: An SDK for next-generation SD-RANs. In *Proceedings of the 17th International Conference on emerging Networking EXperiments and Technologies*. 411–425.

[36] Zewen Shang, Matheus E Garbelini, and Sudipta Chattopadhyay. 2024. U-Fuzz: Stateful Fuzzing of IoT Protocols on COTS Devices. In *ICST*.

[37] Orange Open Source. 2023. ns-O-RAN-flexric: O-RAN and FlexRIC Integration for ns-3. https://github.com/Orange-OpenSource/ns-O-RAN-flexric. Accessed: 2025-08-05.

[38] Software Radio Systems. 2025. O-RAN NearRT-RIC and xApp tutorial. srsRAN Project documentation. https://docs.srsran.com/projects/project/en/latest/tutorials/source/near-rt-ric/source/index.html Accessed: 2025-01-19.

[39] Software Radio Systems. 2025. srsRAN Project: Open-Source 5G CU/DU. GitHub repository. https://github.com/srsran/srsRAN_Project.

[40] Kashyap Thimmaraju et al. 2024. Security testing the o-ran near-real time ric & a1 interface. In *WiSec*. 277–287.

[41] Wireshark Foundation. 2024. Wireshark. https://www.wireshark.org. Network Protocol Analyzer.

[42] Tianchang Yang et al. 2024. {ORANalyst}: Systematic Testing Framework for Open {RAN} Implementations. In *USENIX Security Symposium*. 1921–1938.

[43] Michał Zalewski. 2020. American Fuzzy Lop (AFL): Security-oriented Grey-box Fuzzer. https://github.com/google/AFL. Latest official release 2.57b; uses lightweight instrumentation and genetic mutation for code coverage.

A Appendix

Field Type	Constraints	Mutation Method
Boolean	True, False	Flip the boolean value.
Integer	Range, bounds	Random integer within boundaries.
Enumerated	Valid enums (ASN.1)	Replace with an adjacent enumeration option.
Bit String	Size constraint	Change bit-string length, insert malformed sequences invalid bit patterns.
Octet String	Length constraint	Randomly mutate bytes or insert invalid octet values within the allowed length.
Sequence / Sequence Of	Length, element types	Mutate sequence length or element values.
Choice	Element choice	Switch to another valid choice variant.
Optional	—	Insert or mutate present optional fields.

Table 2: Generation and mutation field types used in *ORANClaw*.

A.1 Genetic Algorithm Optimization

Algorithm 2 Updating population (Genetic Optimizer)

procedure UPDATE($[\mathbf{w}_i, c(\mathbf{w}_i)]_{1\ldots N}$)
  $P_t \leftarrow$ sorted $[\mathbf{w}_i]_{1\ldots N}$ based on decreasing $[c(\mathbf{w}_i)]_{1\ldots N}$
  $\mathbf{w}_A, \mathbf{w}_B \leftarrow P_t[0], P_t[1]$
  $P_{t+1} \leftarrow \{\mathbf{w}_A, \mathbf{w}_B\}$ ▹ Select top two
  while $|P_{t+1}| < N$ **do**
    $\mathbf{w}_c \leftarrow \frac{\mathbf{w}_A + \mathbf{w}_B}{2}$
    $\mathbf{w}_c \leftarrow \frac{\mathbf{w}_c}{\|\mathbf{w}_c\|_1}$ ▹ Crossover + normalize
    if random() $< p_{\text{mut}}$ **then** ▹ Mutation
      $j \sim \text{Uniform}(\{0, 1, 2\})$
      $\mathbf{w}_{c,j} \leftarrow \max(\mathbf{w}_{c,j} + U(-0.2, 0.2), 0.01)$
      $\mathbf{w}_c \leftarrow \frac{\mathbf{w}_c}{\|\mathbf{w}_c\|_1}$ ▹ Re-normalize
  return P_{t+1} ▹ Advance generation

Algorithm 2 outlines the creation of a new population (i.e., $\mathcal{G}$.update in line 13 of Algorithm 1). In particular, we preserve the top two individuals as parents. To create a new individual, we perform arithmetic crossover $\mathbf{w}_c \leftarrow \frac{\mathbf{w}_A + \mathbf{w}_B}{2}$ with L1-normalization $\mathbf{w}_c \leftarrow \frac{\mathbf{w}_c}{|\mathbf{w}_c|_1}$ and preserve convex combination properties of the weights vector $w = [w_{\text{SM}}, w_{\text{json}}, w_{\text{both}}]$. This child is then subject to probabilistic mutation (with probability $p_{\text{mut}} = 0.3$) with uniform perturbation $U(-0.2, 0.2)$ to random components of $\mathbf{w}_c$. This is followed by constraining from below ($\max(\cdot, 0.01)$) and re-normalization to maintain feasible probability space.

B StateMapper Configuration

```
{ "Filter": "e2ap", "StateNameField": "e2ap.procedureCode"},
{ "Filter": "x2ap", "StateNameField": "x2ap.procedureCode"},
{ "Filter": "f1ap", "StateNameField": "f1ap.procedureCode"},
{ "Filter": "e1ap", "StateNameField": "e1ap.procedureCode"},
{ "Filter": "ngap", "StateNameField": "ngap.procedureCode"},
{ "Filter": "sctp", "StateNameField": "sctp.chunk_type"   }
```

Listing 3: StateMapper Rules

Implementation	VulnID	CVE	Vulnerability	General Cause	Component	Threat	Location
O-RAN SC RIC	VulnOSCRIC-1	Pending	Unhandled Exception	Flooding E2SetupRequest	E2 Termination	DoS	Not Specified
OAI	VulnOAI-01	CVE-2024-48408	Assertion	Malformed MAC SM Field	O-DU	DoS	ran_func_mac.c:126
	VulnOAI-02	CVE-2025-52142	Assertion	Malformed KPM SM Field	O-DU/CU-UP	DoS	ran_func_kpm_subs.c:226
	VulnOAI-03	CVE-2025-52146	Assertion	Malformed KPM SM Field	O-DU/CU-UP	DoS	msg_handler_agent.c:136
	VulnOAI-04	CVE-2025-52150	Assertion	**Truncated MAC SM Structure**	O-DU	DoS	mac_dec_plain.c:190
	VulnOAI-05	Pending	Assertion	Malformed KPM SM Field	O-DU	DoS	ran_func_kpm.c:267/268
	VulnOAI-06	Pending	Assertion	**Malformed KPM SM Structure**	O-DU	DoS	ran_func_kpm.c:72
	VulnOAI-07	CVE-2025-52148	Assertion	Unexpected TC SM Field	O-CU-UP	DoS	tc_dec_plain.c:1418
	VulnOAI-08	CVE-2025-52151	Assertion	Malformed KPM SM Field	O-CU-UP	DoS	ran_func_kpm.c:230
	VulnOAI-9	CVE-2025-52147	Assertion	Zero-ed RC SM Field	O-CU-CP	DoS	rc_dec_asn.c:953
	VulnOAI-10	Pending	Assertion	Unable To Recover	O-CU-CP (E1)	DoS	cucp_cuup_e1ap.c:31
	VulnOAI-11	Pending	Assertion	**Malformed KPM SM Structure**	O-CU-UP	DoS	ran_func_kpm.c:177
	VulnOAI-12	Pending	Assertion	Malformed E42RICsubscriptionDeleteRequest Field	O-CU-CP	DoS	bimap.c:126
	VulnOAI-13	Pending	Assertion	Malformed KPM SM Field	O-CU-UP	DoS	ran_func_kpm.c:168
	VulnOAI-14	Pending	Assertion	Multiple Malformed KPM Fields	O-DU/CU-UP	DoS	ran_func_kpm.c:71
	VulnOAI-15	Pending	Assertion	Invalid KPM SM Field	O-CU-CP/CU-UP	DoS	ran_func_kpm.c:229
	VulnOAI-16	Pending	Assertion	Malformed KPM SM Field	O-CU-CP/UP	DoS	ran_func_kpm.c:176
	VulnOAI-17	Pending	Assertion	Malformed KPM SM Field	O-CU-CP/UP	DoS	ran_func_kpm.c:167
	VulnOAI-18	Pending	Assertion	Unable To Recover	O-CU-UP/CP	DoS	e2_agent.c:242
	VulnOAI-19	Pending	Assertion	Invalid KPM SM Field	O-CU-UP/CP	DoS	plugin_agent.c:286
NS-3	VulnNS-01	Pending	Buffer Overflow	Malformed RICSubscriptionRequest Field	gNB	Mem. Corrupt	Not Specified
	VulnNS-02	Pending	Heap-based Buffer Overflow	Malformed E42SetupRequest Field	gNB	Mem. Corrupt	Not Specified
	VulnNS-03	Pending	Heap-Based Buffer Overflow	E42SetupRequest Duplication	gNB	Mem. Corrupt	Not Specified
	VulnNS-04	Pending	Assertion	Malformed E42RICSubscriptionDeleteRequest	gNB	DoS	ipv4-l3-protocol.cc:972
	VulnNS-05	Pending	Assertion	Malformed RICSubscriptionRequest Field	gNB	DoS	ipv4-l3-protocol.cc:580
	VulnNS-06	Pending	Assertion	Malformed RICSubscriptionRequest Field	LTE eNB	DoS	lte-spectrum-phy.cc:486
	VulnNS-07	Pending	Null pointer dereference	Malformed RICSubscriptionRequest Field	gNB	Mem. Corrupt	ptr.h:638
	VulnNS-08	Pending	Out-of-bonds read	Malformed RICSubscriptionRequest Field	gNB	Info. Disclosure	net-device-queue-interface.cc:216
	VulnNS-09	Pending	Null pointer dereference	Malformed RICSubscriptionRequest Field	gNB	Mem. Corrupt	ptr.h:630
	VulnNS-10	Pending	Assertion	Malformed RICSubscriptionRequest Field	gNB	DoS	object.cc:349
	VulnNS-11	Pending	Assertion	Malformed RICSubscriptionRequest Field	gNB	DoS	traffic-control-layer.cc:337
	VulnNS-12	Pending	Improper Input Validation	Malformed E42RICSubscriptionDeleteRequest Field	gNB	DoS	point-to-point-net-device.cc:279
	VulnNS-13	Pending	Improper Check for Unusual or Exceptional Conditions	Malformed E42RICSubscriptionDeleteRequest Field	gNB	Improper File Handling	mmwave-phy-trace.cc:399
	VulnNS-14	Pending	Assertion	Invalid KPM RC Field	gNB	DoS	default-simulator--impl.cc:235
	VulnNS-15	Pending	Assertion	Malformed RIC control Request Field	gNB	DoS	buffer.cc:183
	VulnNS-16	Pending	Improper Input Validation	Malformed E42RICSubscriptionDeleteRequest Field	gNB	DoS	mmwave-phy-trace.cc:220
	VulnNS-17	Pending	Improper Check for Unusual or Exceptional Conditions	Malformed E42RICSubscriptionDeleteRequest Field	gNB	Improper File Handling	mmwave-phy-trace.cc:71
	VulnNS-18	Pending	Assertion	Malformed RICSubscriptionRequest Field	LTE eNB	Dos	mmwave-enb-phy.cc:1141
FlexRIC	-	CVE-2024-34034 (Existing)	Assertion	Foloding E42SubscriptionRequest	RIC	DoS	Not Specified
	VulnFlex-01	Pending	Assertion	Malformed RICSubscriptionRequest Field	RIC	DoS	msg_handler_iapp.c:343
	VulnFlex-02	Pending	Assertion	Malformed E42RICsubscriptionDelete Fields	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:2534
	VulnFlex-03	Pending	Assertion	Malformed E42SubscriptionRequest Fields	RIC	DoS	msg_handler_ric.c:117
	VulnFlex-04	Pending	Assertion	Malformed E42SubscriptionRequest Fields	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:544
	VulnFlex-05	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:487
	VulnFlex-06	Pending	Assertion	Malformed MAC SM Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:1107
	VulnFlex-07	Pending	Assertion	Malformed E42SubscriptionDelete Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:2523
	VulnFlex-08	Pending	Assertion	Malformed E42SubscriptionDelete Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:2540
	VulnFlex-09	Pending	Assertion	Flooding E42SetupRequest	RIC	DoS	e2ap_msg_enc_asn.c:3165
	VulnFlex-10	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:536
	VulnFlex-11	Pending	Assertion	Malformed E42SubscriptionDeleteRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:2531
	VulnFlex-12	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:527
	VulnFlex-13	Pending	Assertion	E42SubscriptionDeleteRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:2548
	VulnFlex-14	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v2.03)	DoS	e2ap_msg_dec_asn.c:477
	VulnFlex-15	Pending	Assertion	Unable to Recover	RIC	DoS	map_e2_node_sockaddr.c:154
	VulnFlex-16	Pending	Assertion	Malformed KPM SM Field	RIC	DoS	reg_e2_nodes.c:174
	VulnFlex-17	Pending	Assertion	Unable to Recover	RIC	DoS	map_ric_id.c:227
	VulnFlex-18	Pending	Assertion	Unable to Recover	RIC	DoS	assoc_rb_tree.c:457
	VulnFlex-19	Pending	Assertion	Flooding E42SubscriptionRequest	RIC	DoS	msg_handler_iapp.c:342
	VulnFlex-20	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:418
	VulnFlex-21	Pending	Assertion	E42SubscriptionDeleteRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:435
	VulnFlex-22	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:378
	VulnFlex-23	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:368
	VulnFlex-24	Pending	Assertion	Malformed E42SetupRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:2113
	VulnFlex-25	Pending	Assertion	Unable To Recover	RIC E2AP(v1.01)	DoS	e2ap_msg_enc_asn.c:2731
	VulnFlex-26	Pending	Assertion	Malformed E42SubscriptionRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:427
	VulnFlex-27	Pending	Assertion	Malformed E42SetupRequest Field	RIC E2AP(v1.01)	DoS	e2ap_msg_dec_asn.c:2101
	VulnFlex-28	Pending	Assertion	Unable To Recover	RIC	DoS	endpoint_ric.c:64

Table 3: Vulnerabilities Discovered by *ORANClaw*. Entries marked in bold under *General Cause* indicate vulnerabilities resulting from structural message mutations rather than single-field modifications. "*Unable To Recover*" denotes failures to restore normal operation of the Component from crashes in previous sessions.